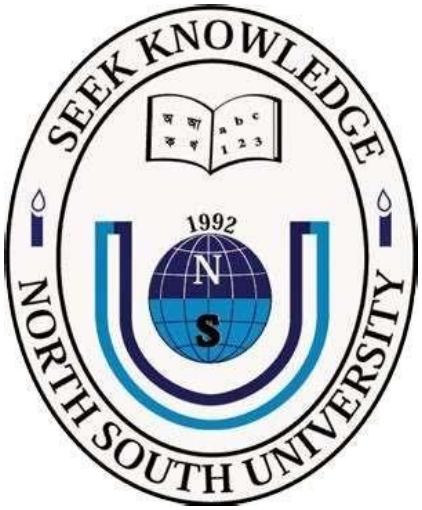
Analog Electronics II



Course instructor:

Dr. Mohsin Sajjad

Assistant Professor

North South University

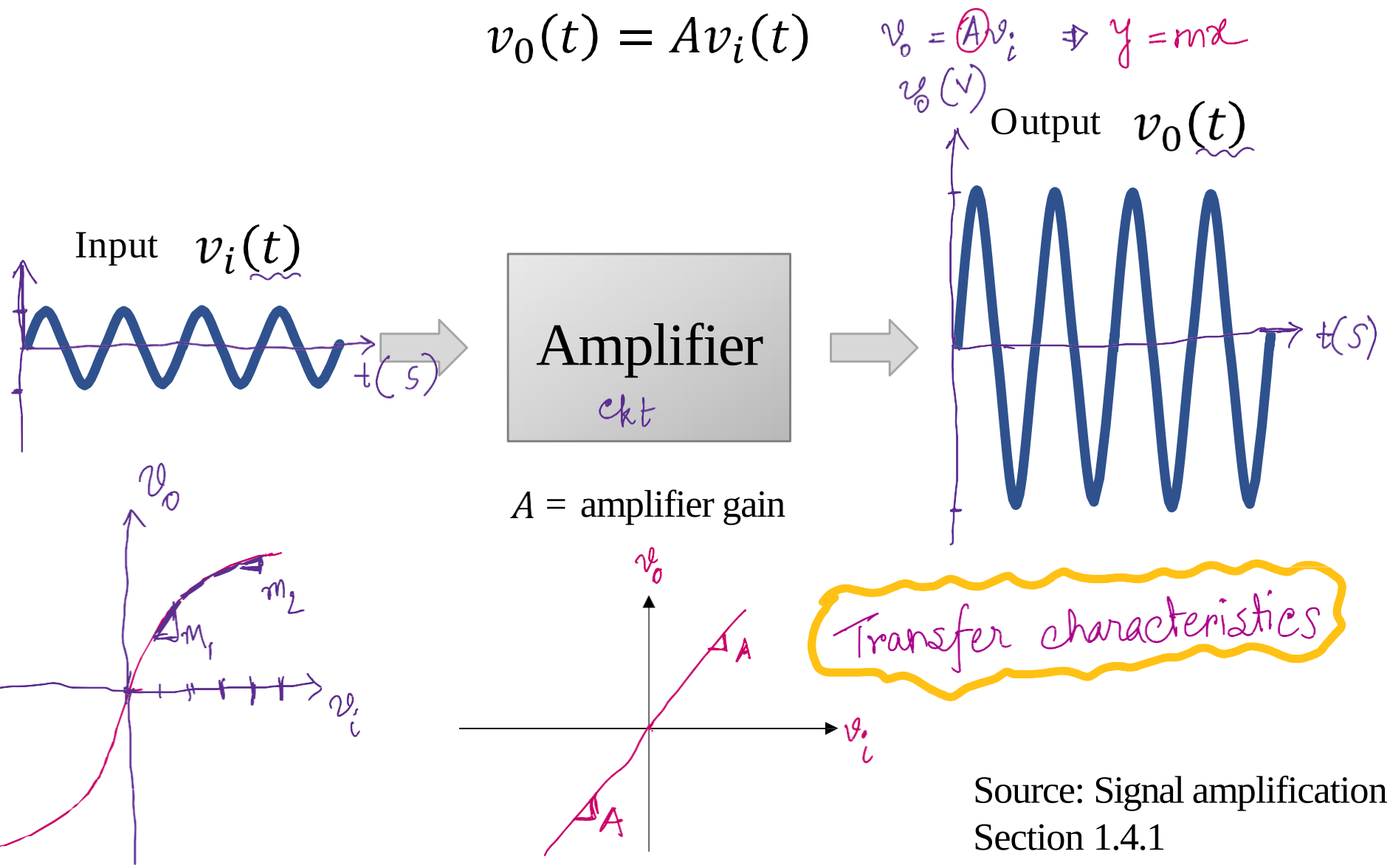
EEE 311

Lecture # 3

Amplifier

For linear amplifier,

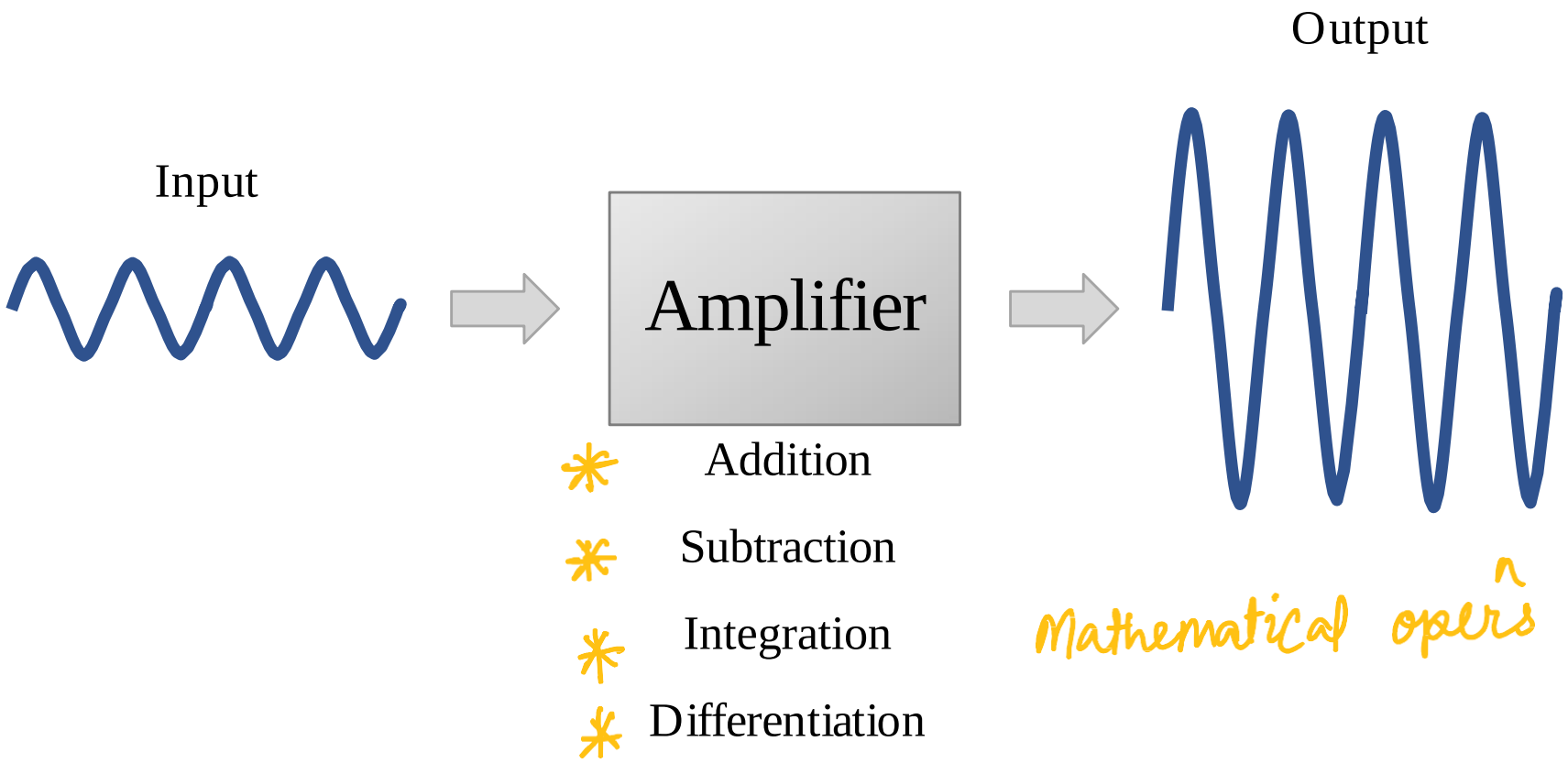
$$v_0(t) = Av_i(t)$$



Source: Signal amplification
Section 1.4.1
Sedra Smith

OP-AMP: an amplifier

OP-AMP: Operational amplifier



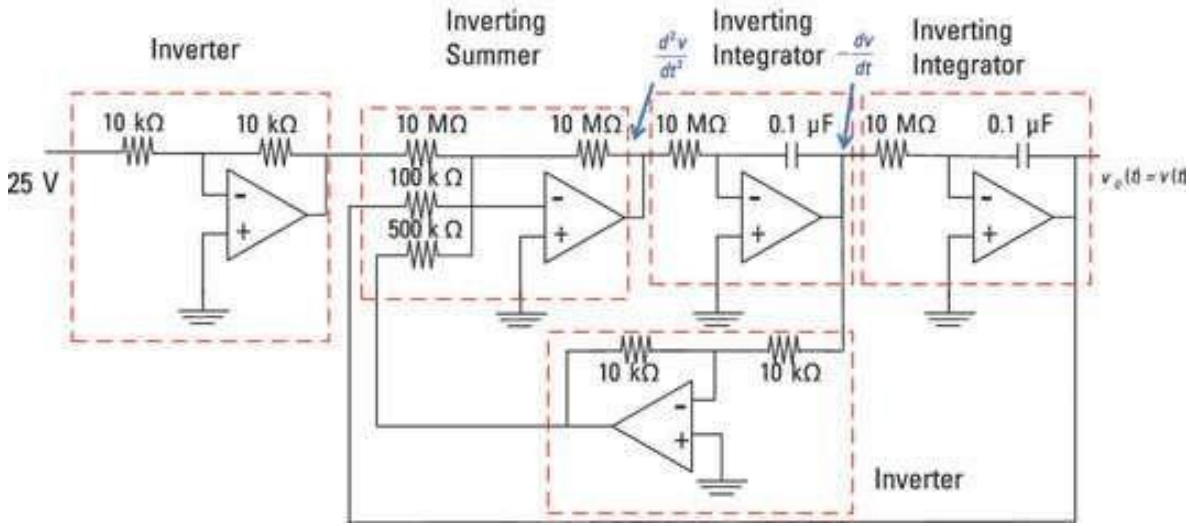
Operational amplifier or Op-Amp

- One of the most versatile and widely used electronic devices in linear applications $v_o \propto v_i$ (linear)
- Low-cost and easy-to-use
- Many operations, such as addition, subtraction, multiplication, integration, differentiation, solving differential equations, and so on

$v_o = 2 \cdot v_1 + 3 \cdot v_2 - 8 \cdot v_3$

$y'', y', y \rightarrow x$

$$\frac{d^2v}{dt^2} = -20\frac{dv}{dt} - 100\,v + 25\text{ V}$$



$v_o = v_1 + v_2$

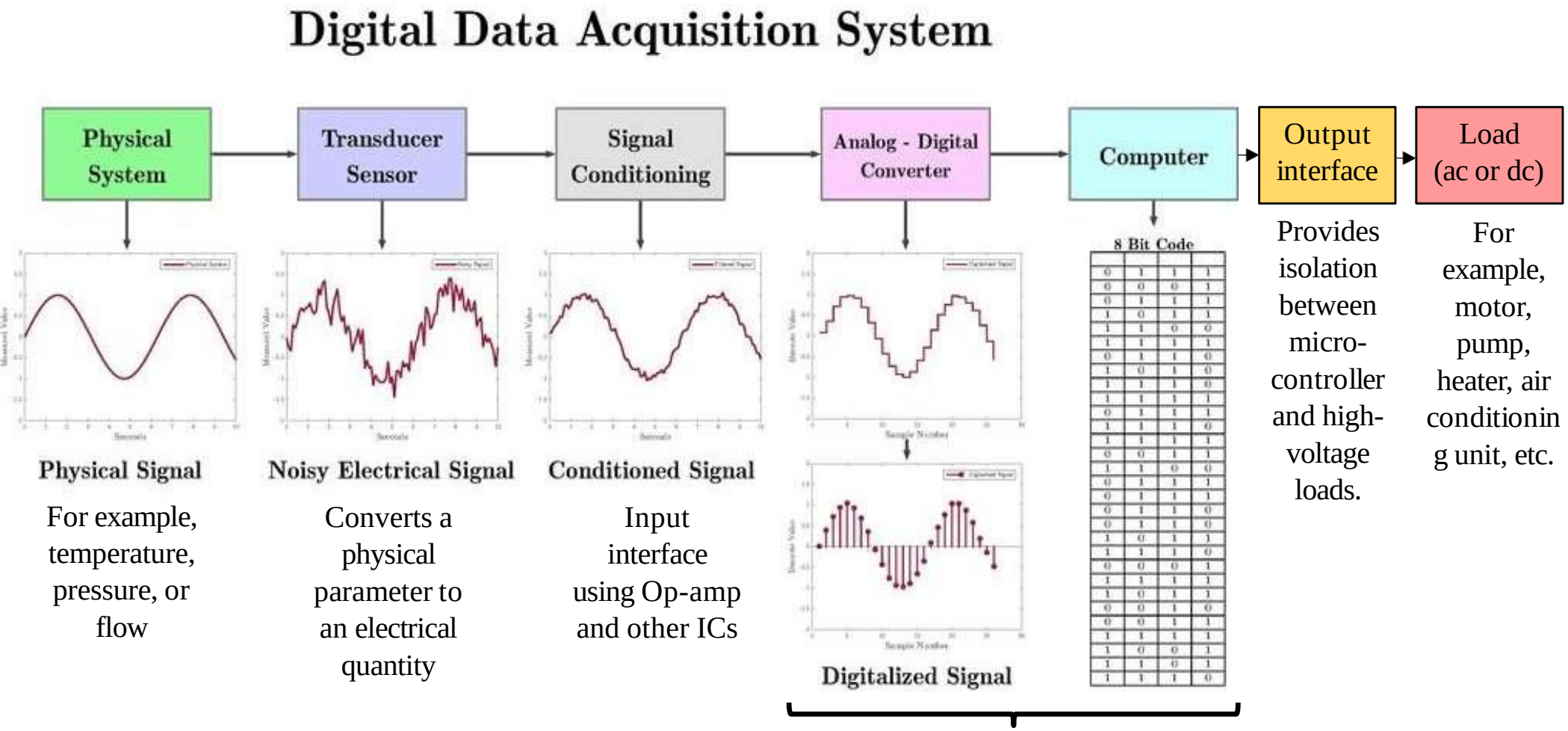
$v_o = v_1 - v_2$

$v_o = 4 \cdot v_1 + 5 \cdot v_2$

$v_o = 2 \cdot v_1 - 8 v_2$

Source: Coughlin, section 1-0

Why analog circuit in the digital world?



For example, temperature, pressure, or flow

Converts a physical parameter to an electrical quantity

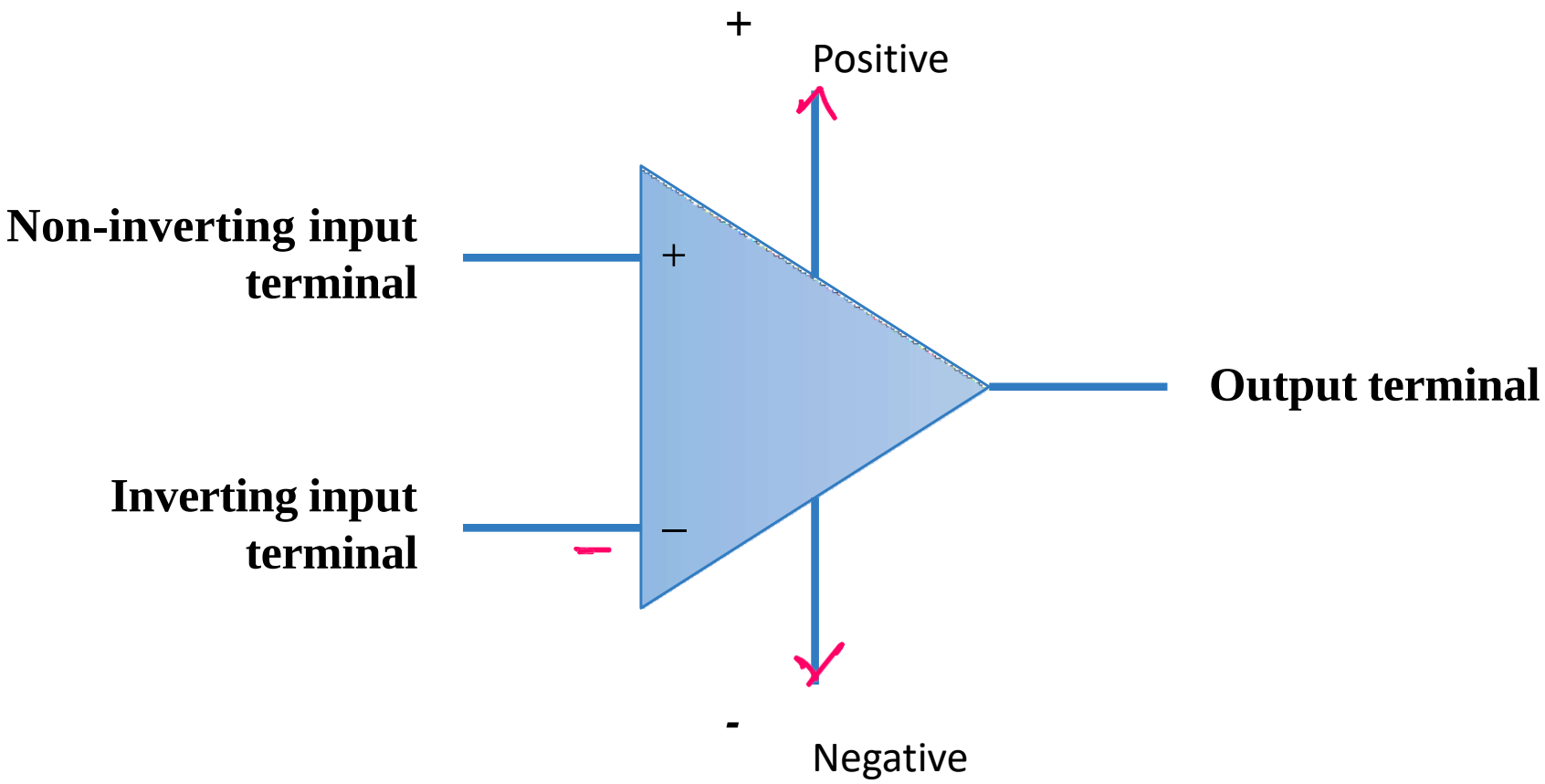
Input interface using Op-amp and other ICs

Provides isolation between micro-controller and high-voltage loads.

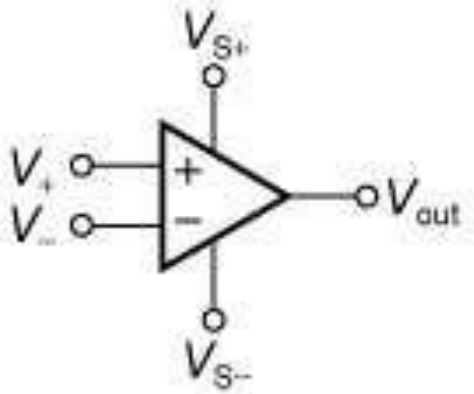
For example, motor, pump, heater, air conditionin g unit, etc.

Source: Coughlin, section 1-1 and Wikipedia

OP-AMP circuit symbol



Circuit symbol



(a)

Dual-in-line package (DIP)

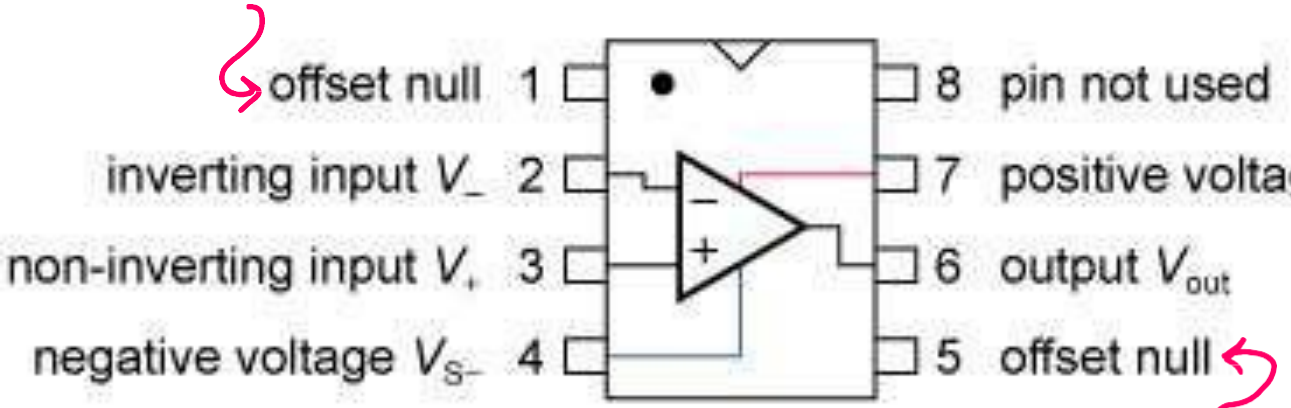


Notch
Or dot

Plastic / ceramic case

(b) Terminals are numbered counter-clockwise

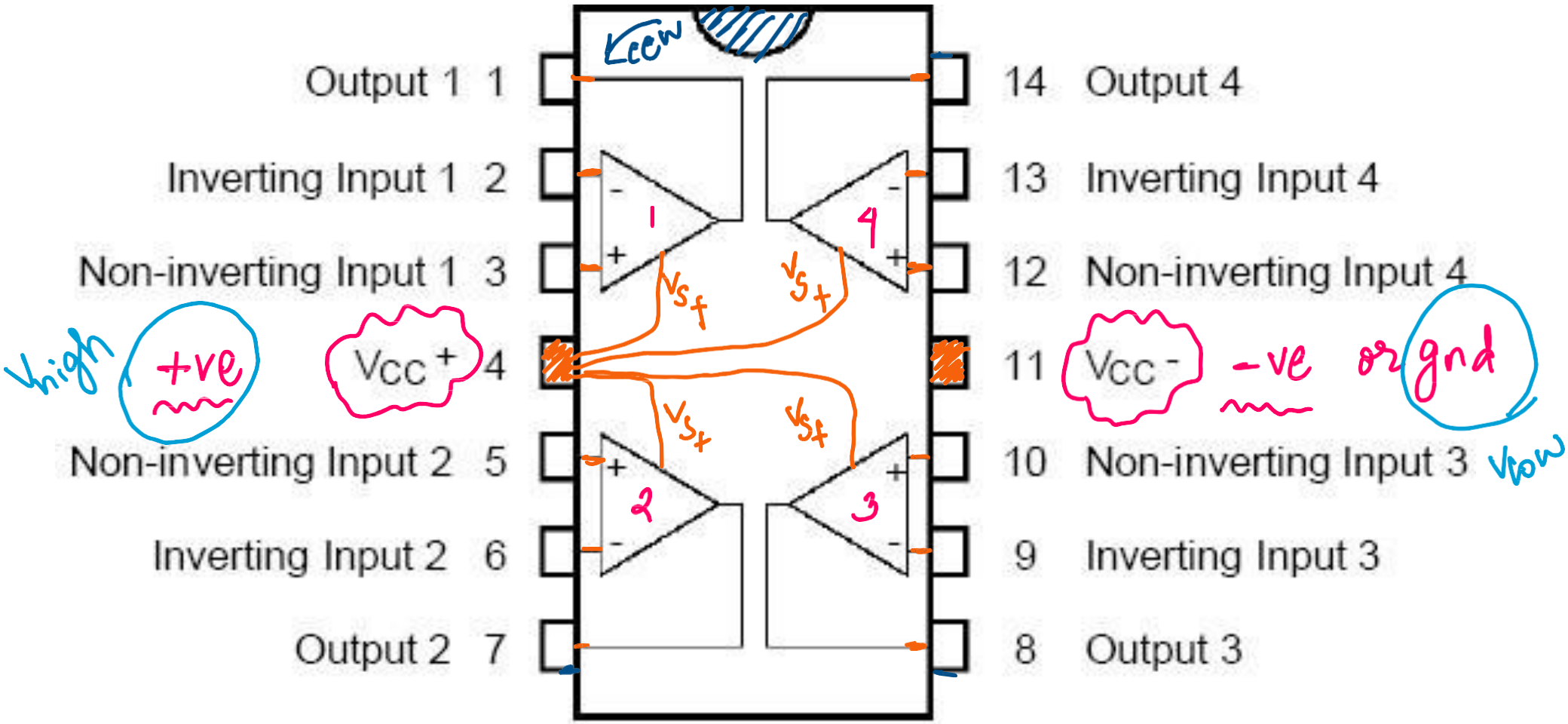
Notch
Or dot



(c) Connection diagram

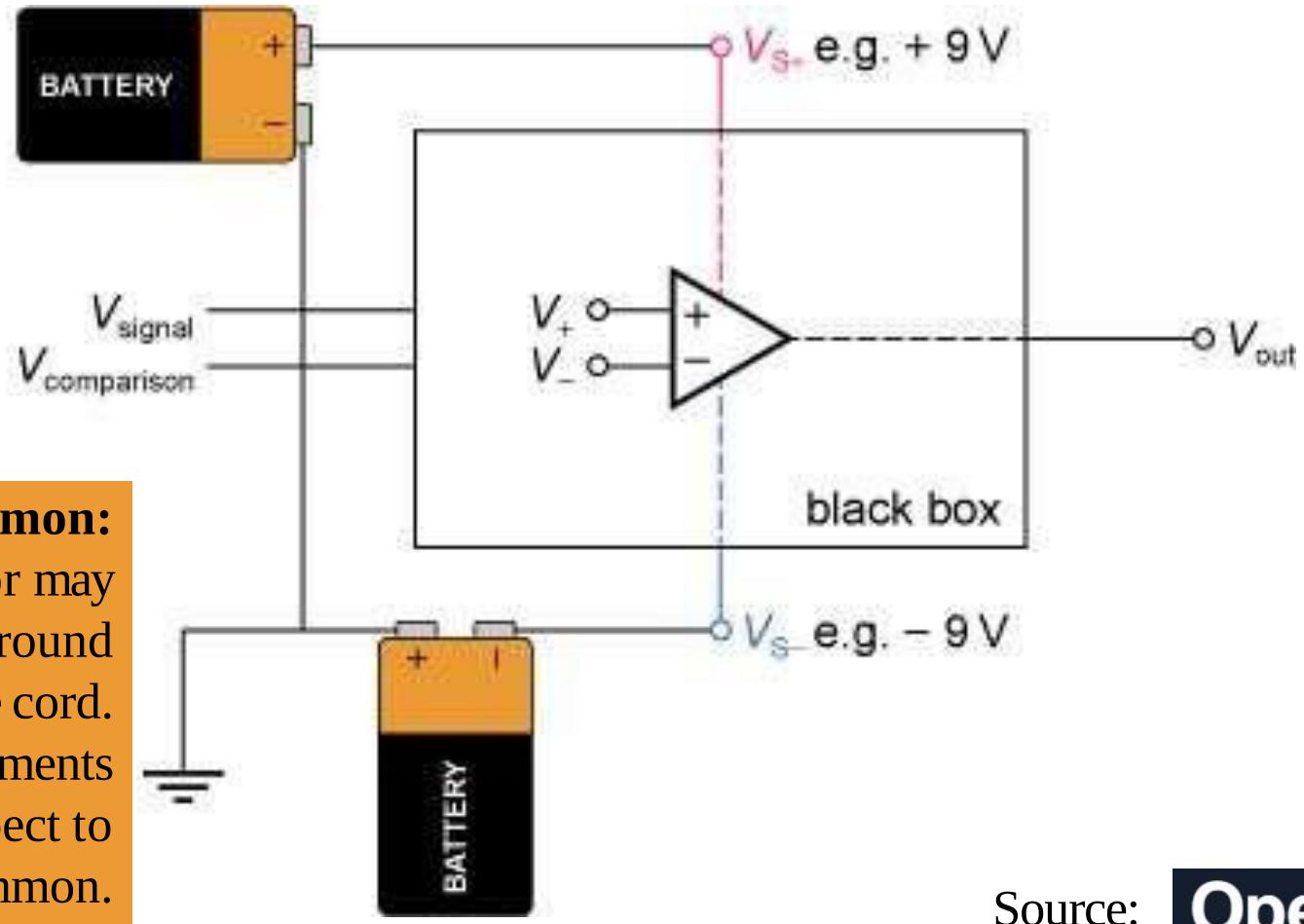
Source: OpenLearn

Quad op-amp



Source: Soundtronics.co.uk

Dual power supplies



Power supply common: This terminal may or may not be wired to earth ground via the 3rd wire of line cord. All voltage measurements are made with respect to power supply common.

Source: OpenLearn

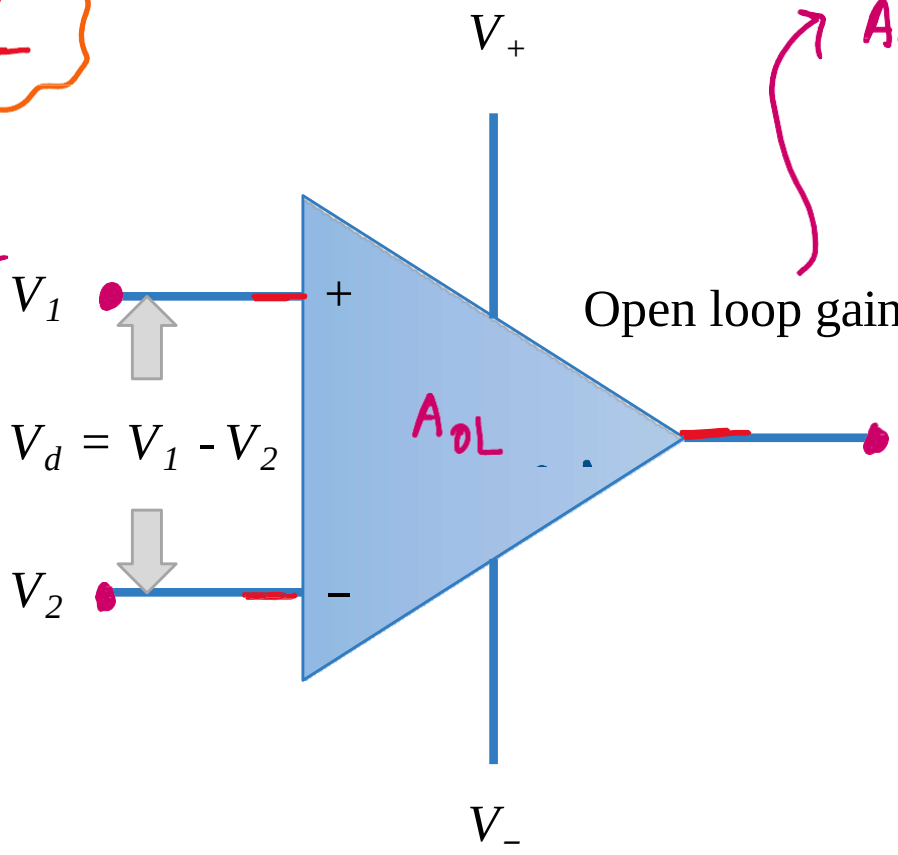
Coughlin 1-6.1, 2-1.1

OP-AMP: a differential amplifier

Op-amp senses the difference between the voltage signals applied at its two input terminals.

$V_d = V_+ - V_-$

$V_d = V_1 - V_2$



practically $A_{OL} = 10^{5-6}$

ideally $A_{OL} = \infty$

Open loop gain = A_{OL}

$V_{out} = A_{OL} \times (V_1 - V_2)$

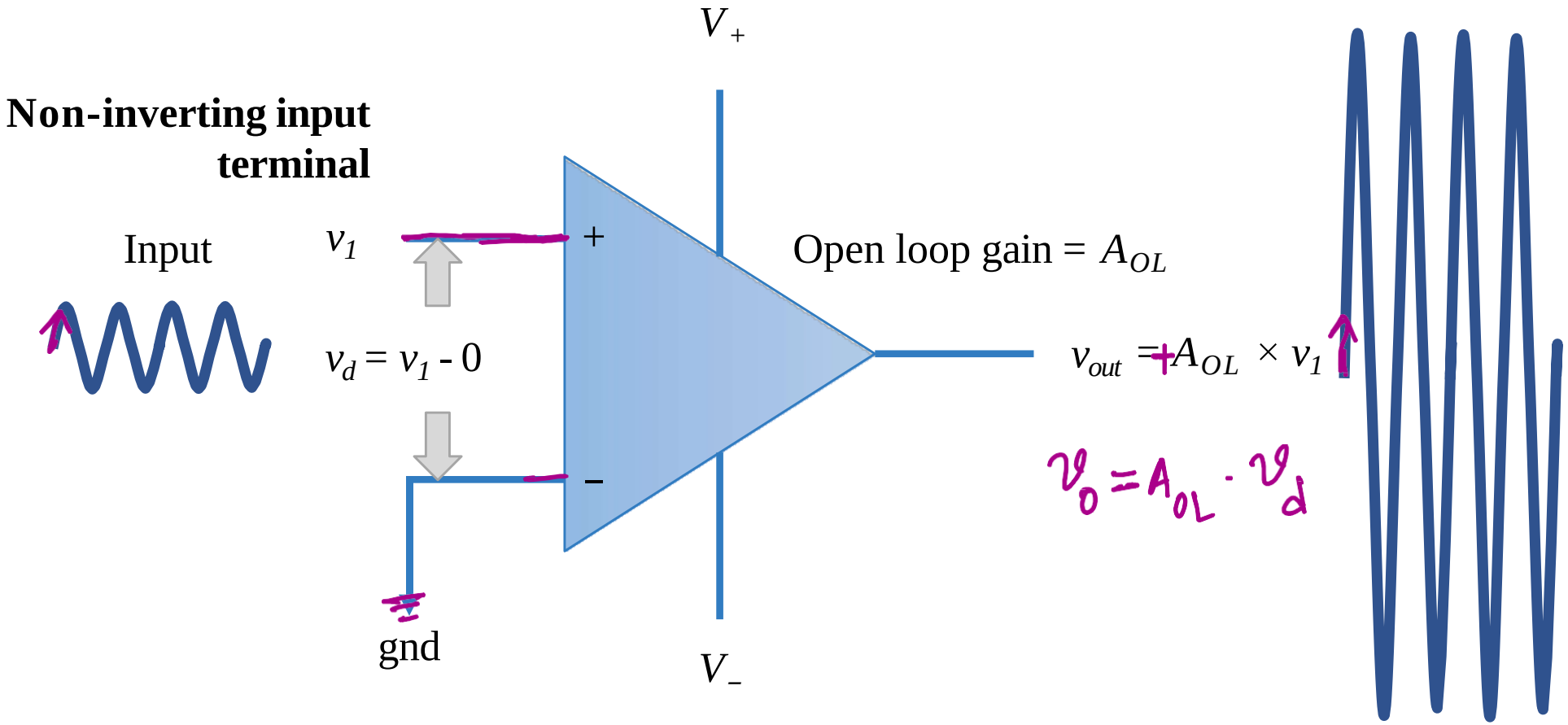
$= A_{OL} V_d$

open-loop configuration : No feedback

OP-AMP: a differential amplifier

$v_d = v_+ - v_- = v_1 - 0V = v_1$

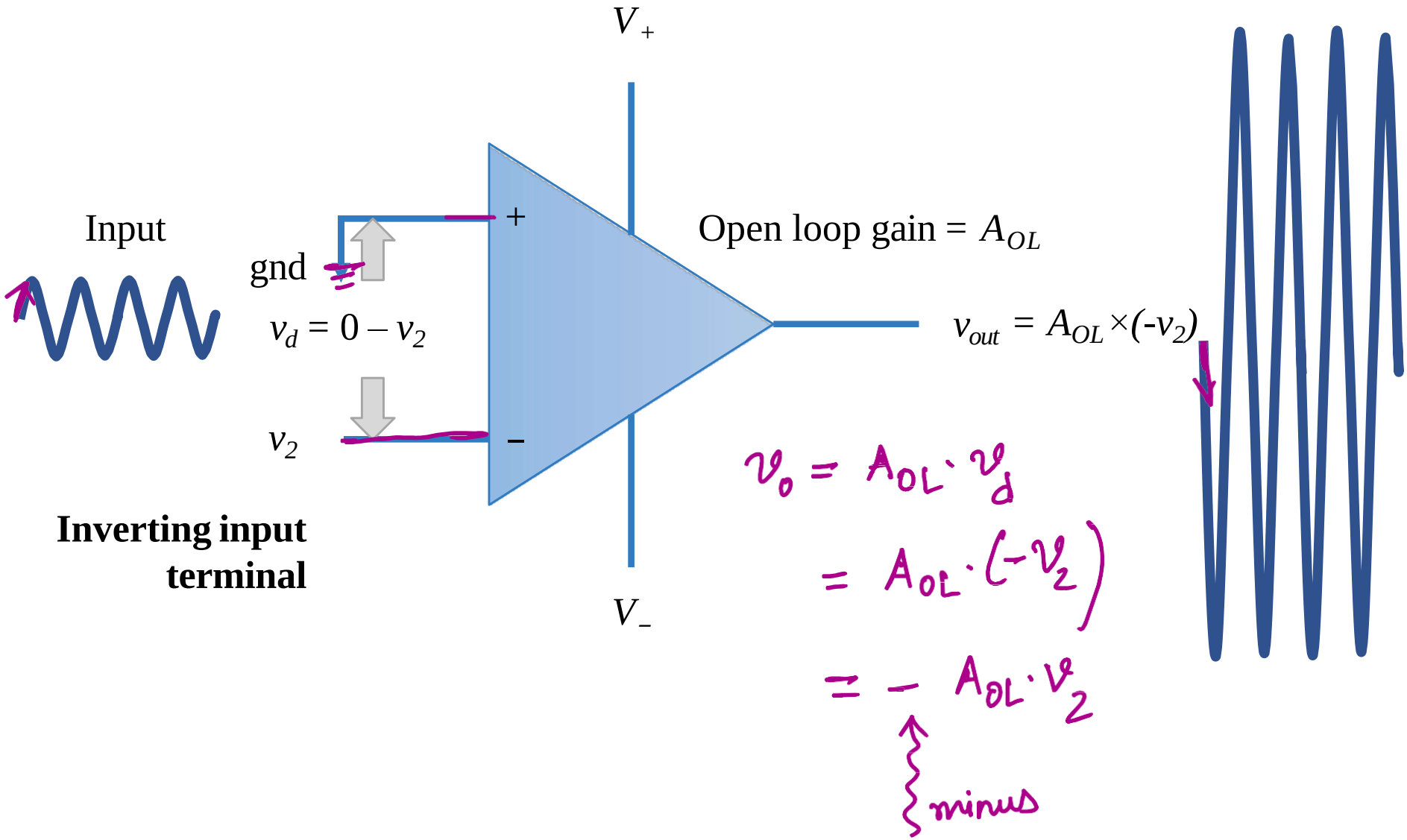
Output: Same phase



OP-AMP: a differential amplifier

$v_d \quad v_+ - v_- = 0V - v_2 = -v_2$

Output: 180° phase



$v_o = A_{OL} \cdot v_d$

$= A_{OL} \cdot (-v_2)$

$= - A_{OL} \cdot v_2$

minus

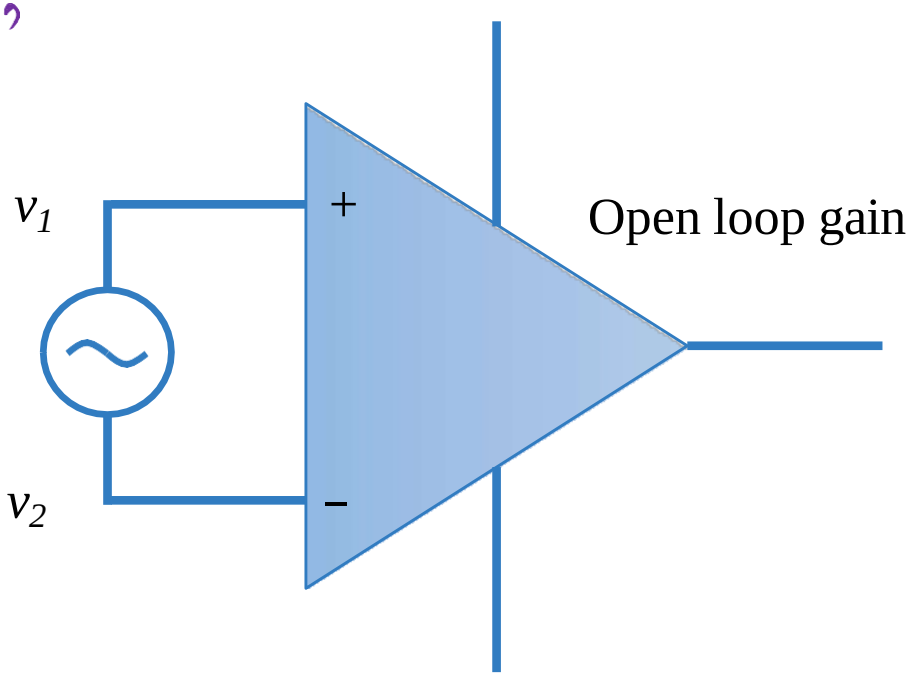
OP-AMP: a differential amplifier

Linear $v_d = 1\,\mu V = 10^{-6}\,V$

Linear $v_d = 10\,\mu V = 10^{-5}\,V$,

$v_d = 100\,\mu V = 10^{-4}\,V$,

Saturation $= 0.1\,mV$



Open loop gain = A_{OL} = 10^5 - 10^6

Saturation $v_d = v_1 - v_2$

= 1 mV

$v_{out} = A_{OL} \times (v_d)$

= 100 V to 1 kV
(theoretically)
(but not practically)
V_+

Saturation $v_d = 10\,mV$

Saturation $v_d = 100\,mV$

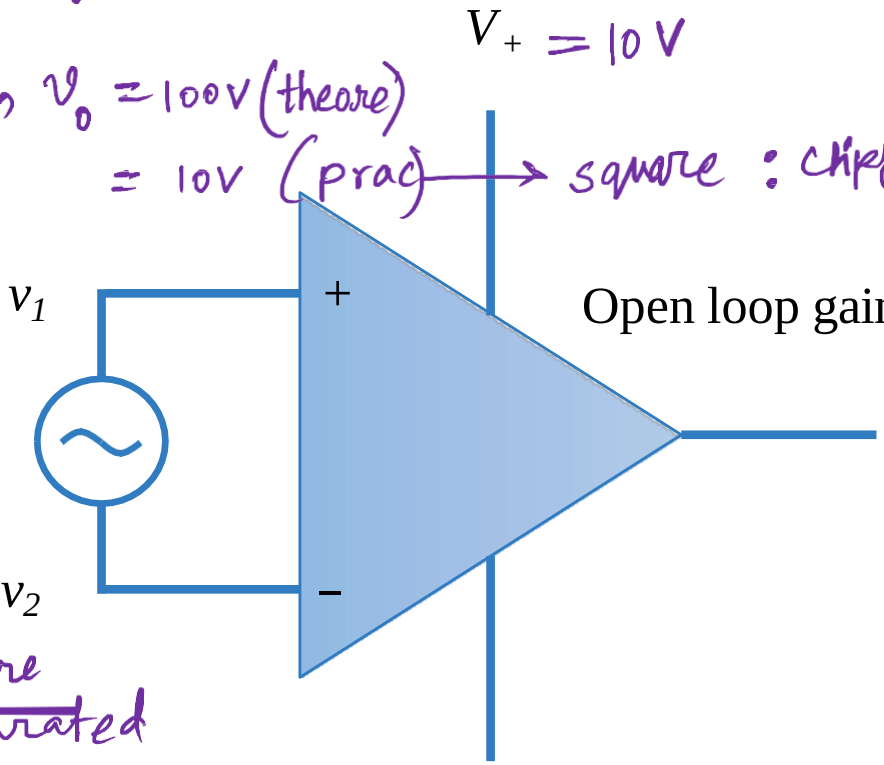
Saturation $v_d = 1\,V$

OP-AMP: a differential amplifier

Linear $v_d = 1\,\mu V = 10^{-6}\,V$, $v_o = 1\,V$

Linear $v_d = 10\,\mu V = 10^{-5}\,V$, $v_o = 10\,V$ } (both theore & prac) i/p & o/p waveforms will look similare

Saturation $v_d = 100\,\mu V = 10^{-4}\,V$, $v_o = 100\,V$ (theore)
$= 0.1\,mV$ $= 10\,V$ (prac) → square : clipped / saturated



Saturation $v_d = v_1 - v_2$

$= 1$ mV

v_o = Square saturated

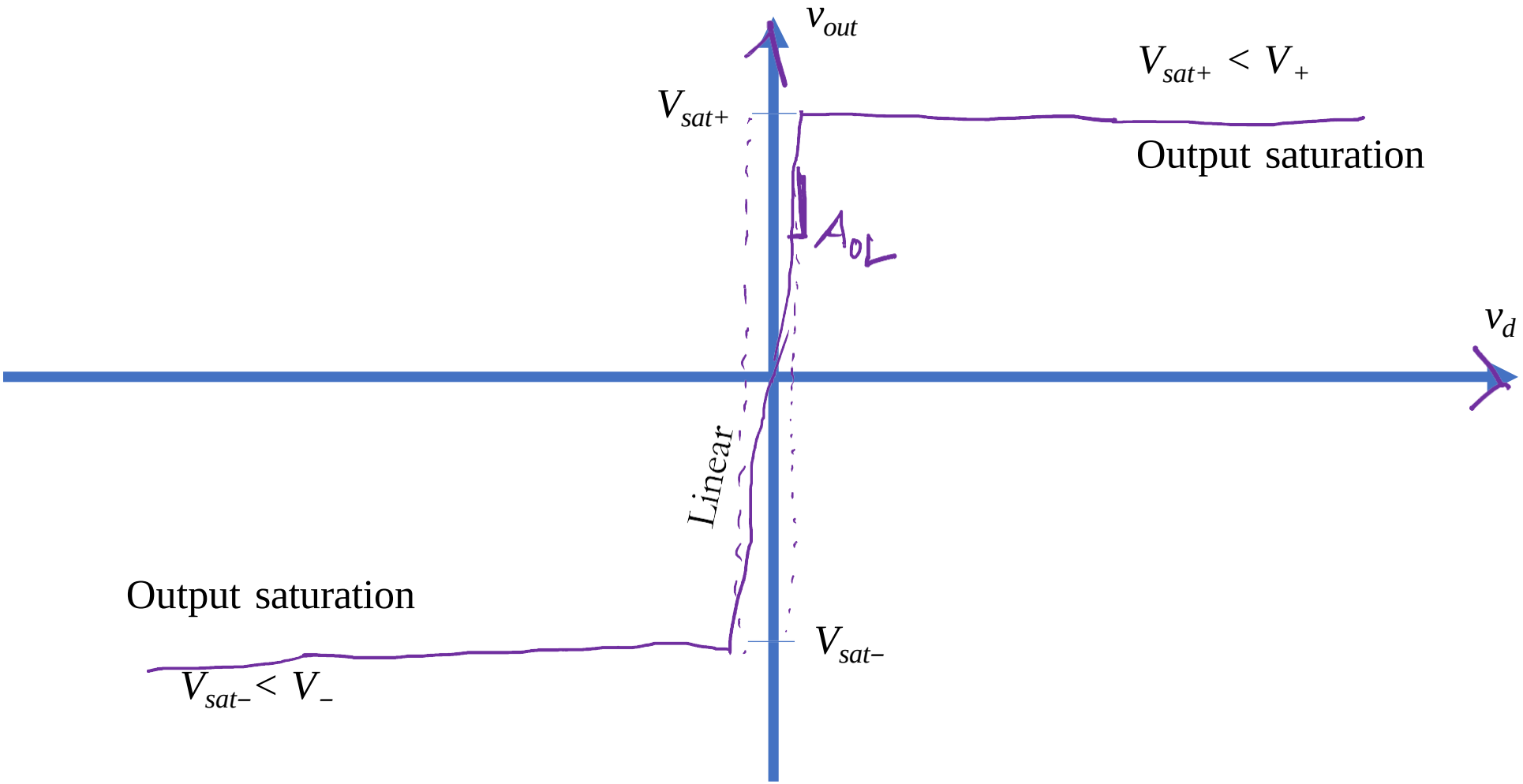
= 100 V to 1 kV
(theoretically)
(but not practically)
V_+

Saturation $v_d = 10$ mV → v_o saturated

Saturation $v_d = 100$ mV ,,

Saturation $v_d = 1\,V$,,

OP-AMP: transfer characteristics

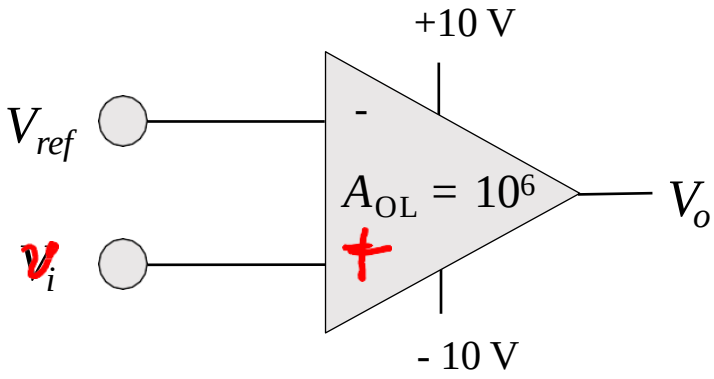


OP-AMP applications

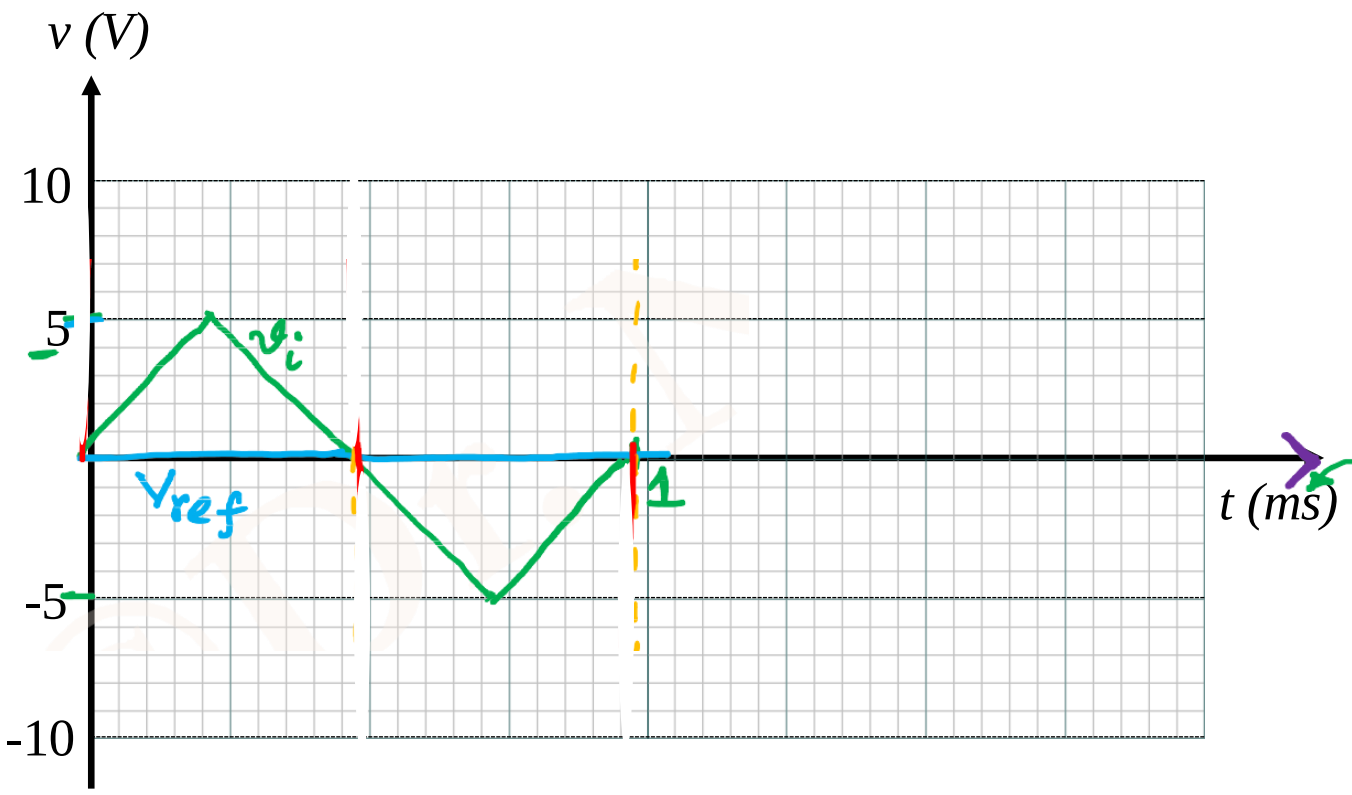
- Adder, subtractor, differentiator, integrator, etc.
- Comparator
- Active filter
- Oscillator
- Waveform convertor
- Analog-to-digital converter & Digital-to-analog convertor

Non-inverting zero crossing detector

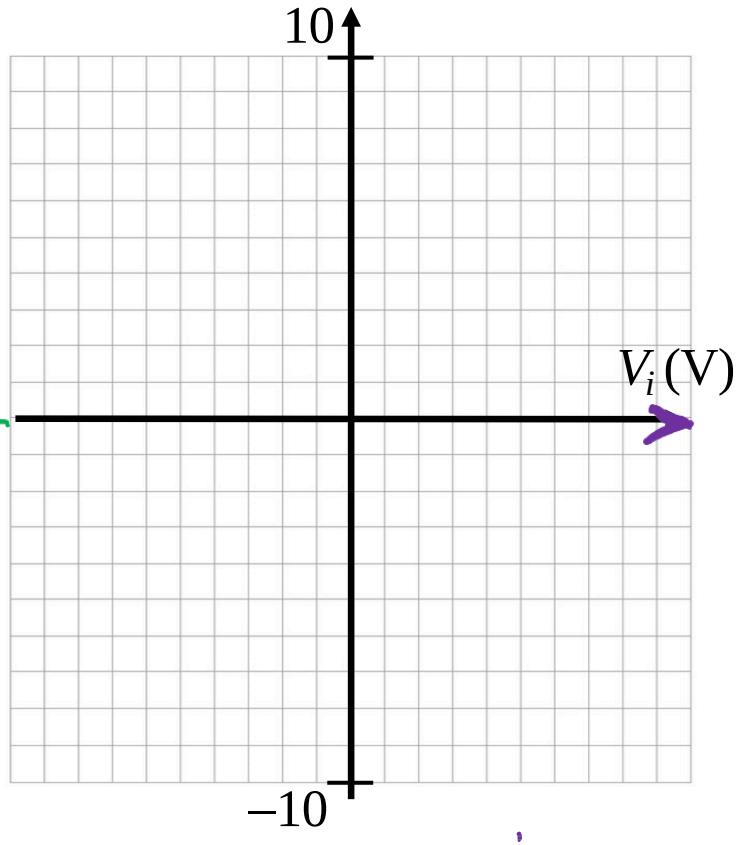
Problem 1. The input signal v_i is a triangular waveform of 1 kHz frequency and 5 V peak. The reference V_{ref} is 0 V. Draw (a) the input waveform v_i, (b) the output waveform v_o and (c) the transfer characteristics.



(a, b)

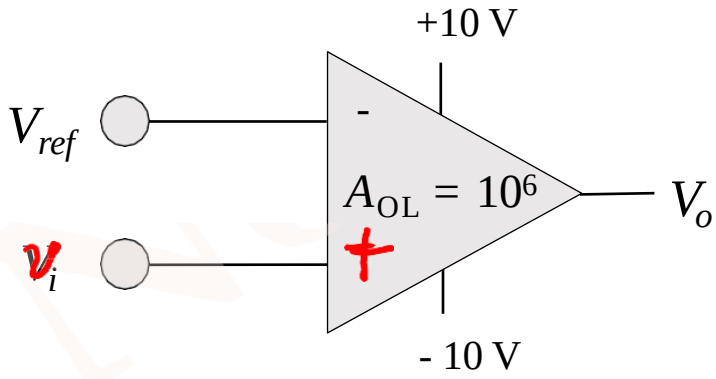


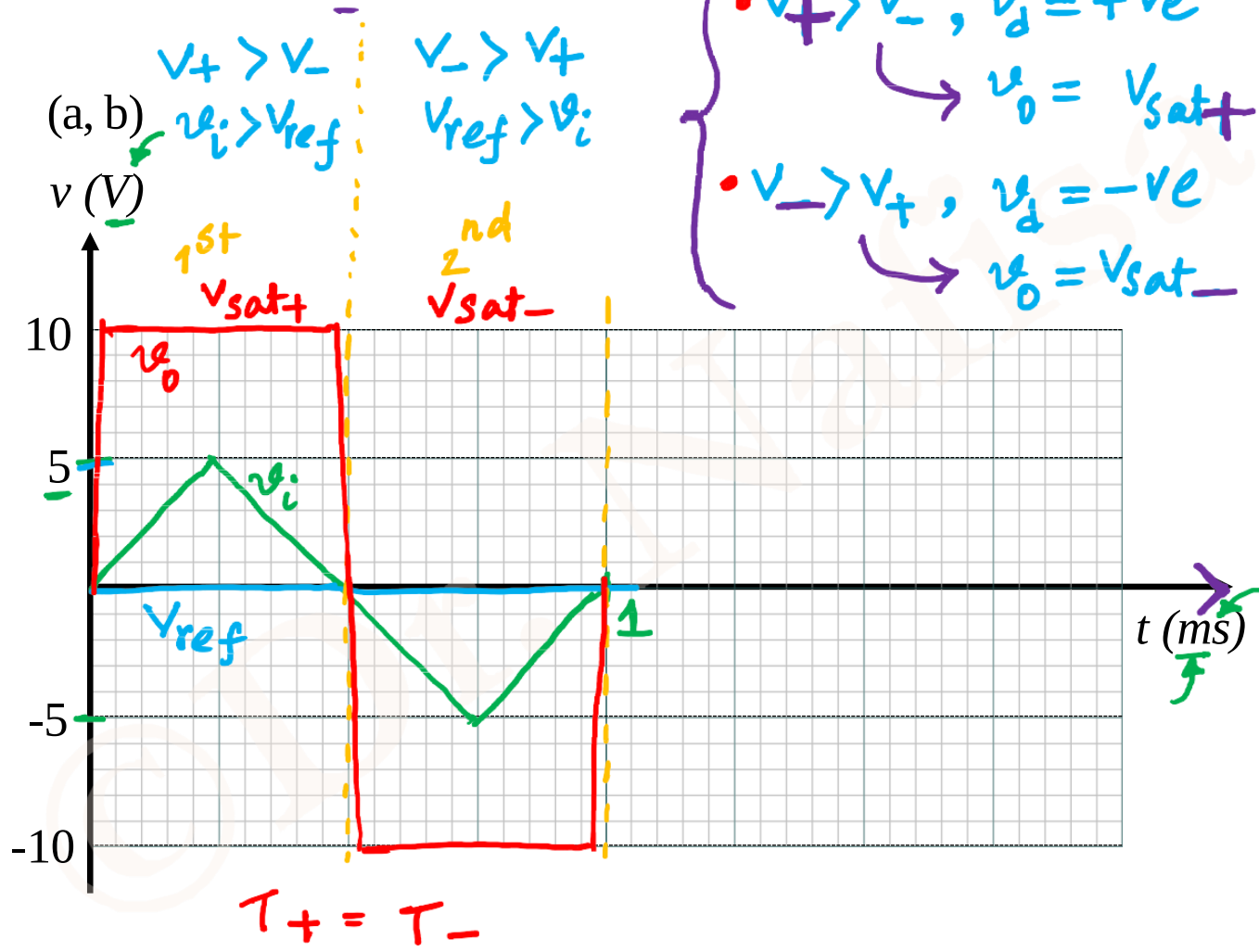
(c) V_O (V)

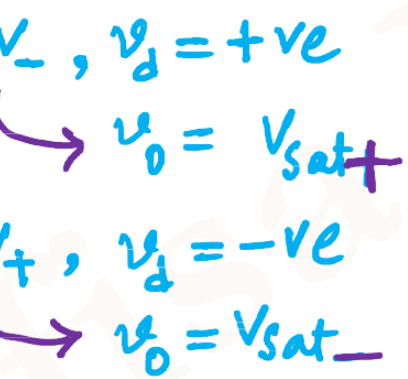


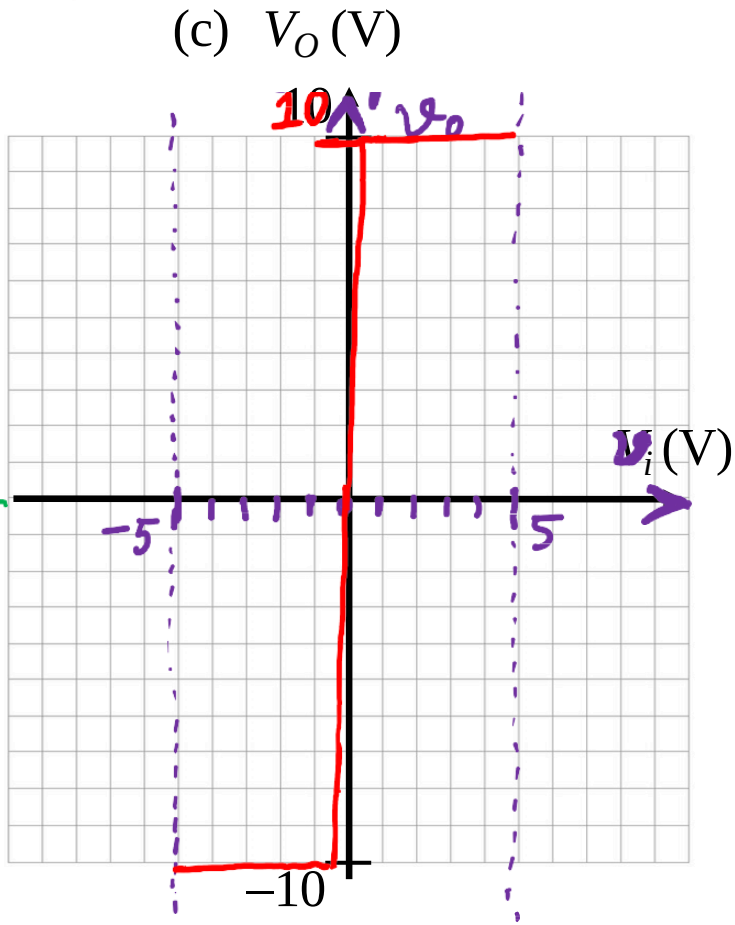
Non-inverting zero crossing detector

Problem 1. The input signal v_i is a triangular waveform of 1 kHz frequency and 5 V peak. The reference V_{ref} is 0 V. Draw (a) the input waveform v_i, (b) the output waveform v_o and (c) the transfer characteristics.



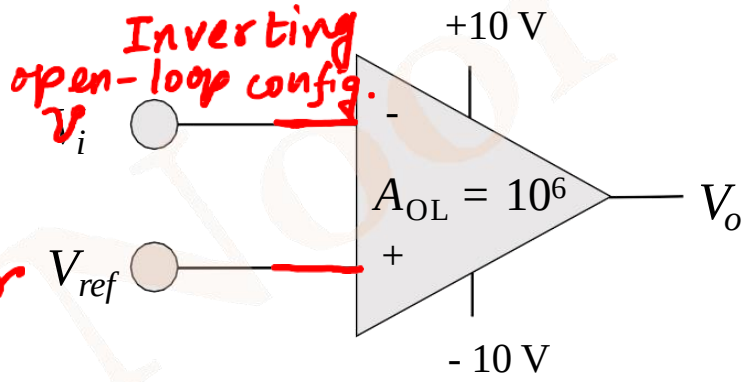




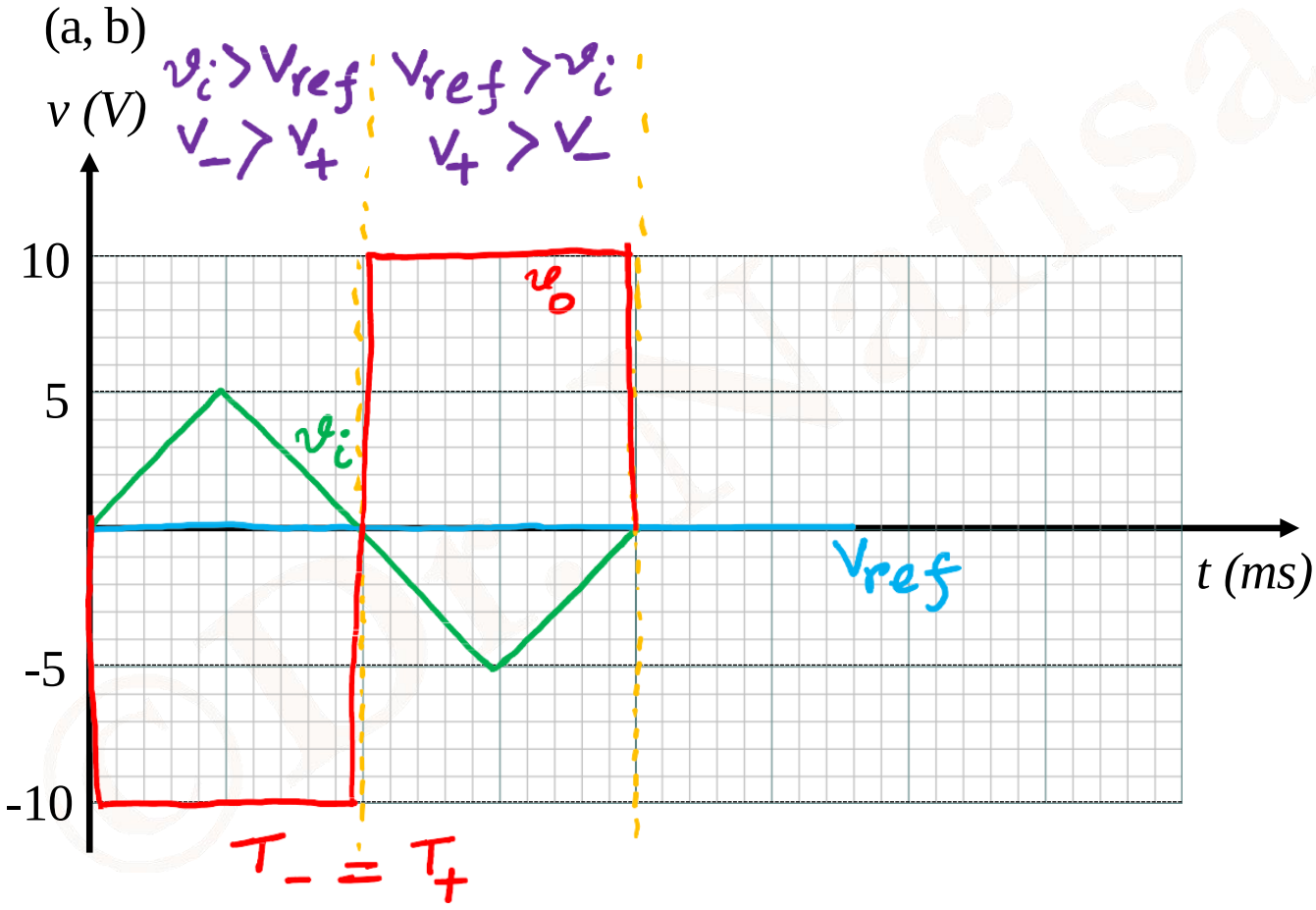


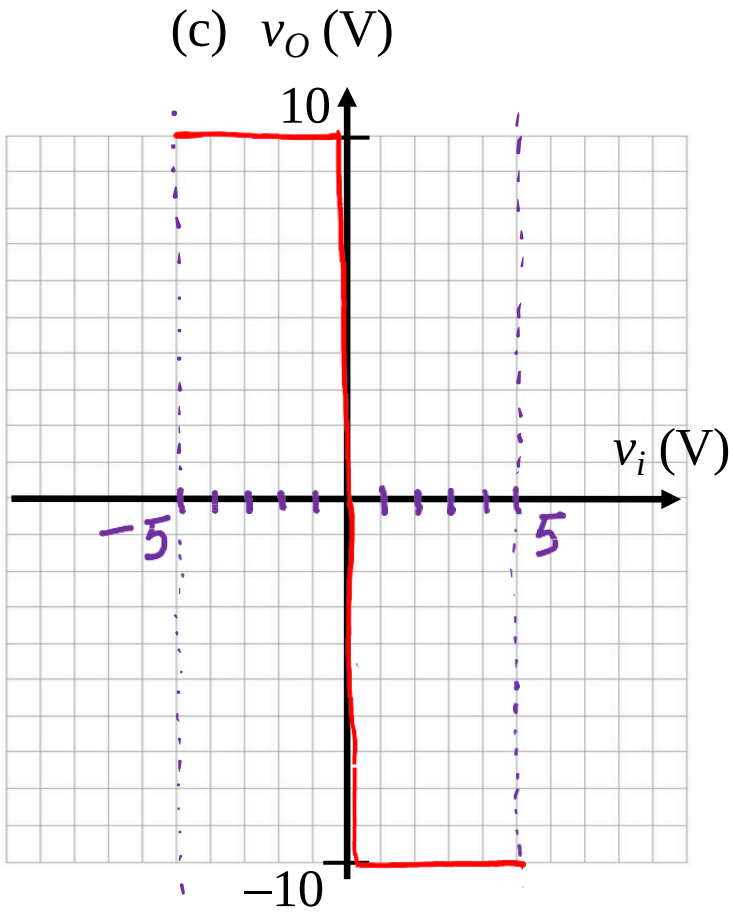
Inverting zero crossing detector

Problem 2. The input signal v_i is a triangular waveform of 1 kHz frequency and 5 V peak. The reference V_{ref} is 0 V. Draw (a) the input waveform v_i, (b) the output waveform v_o and (c) the transfer characteristics.



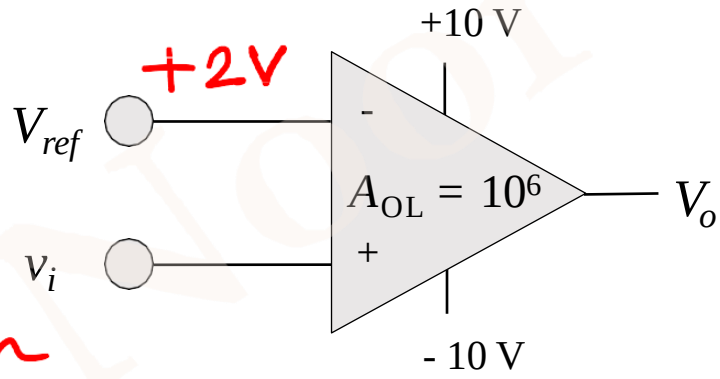
open-loop comparator





Non-inverting positive level detector

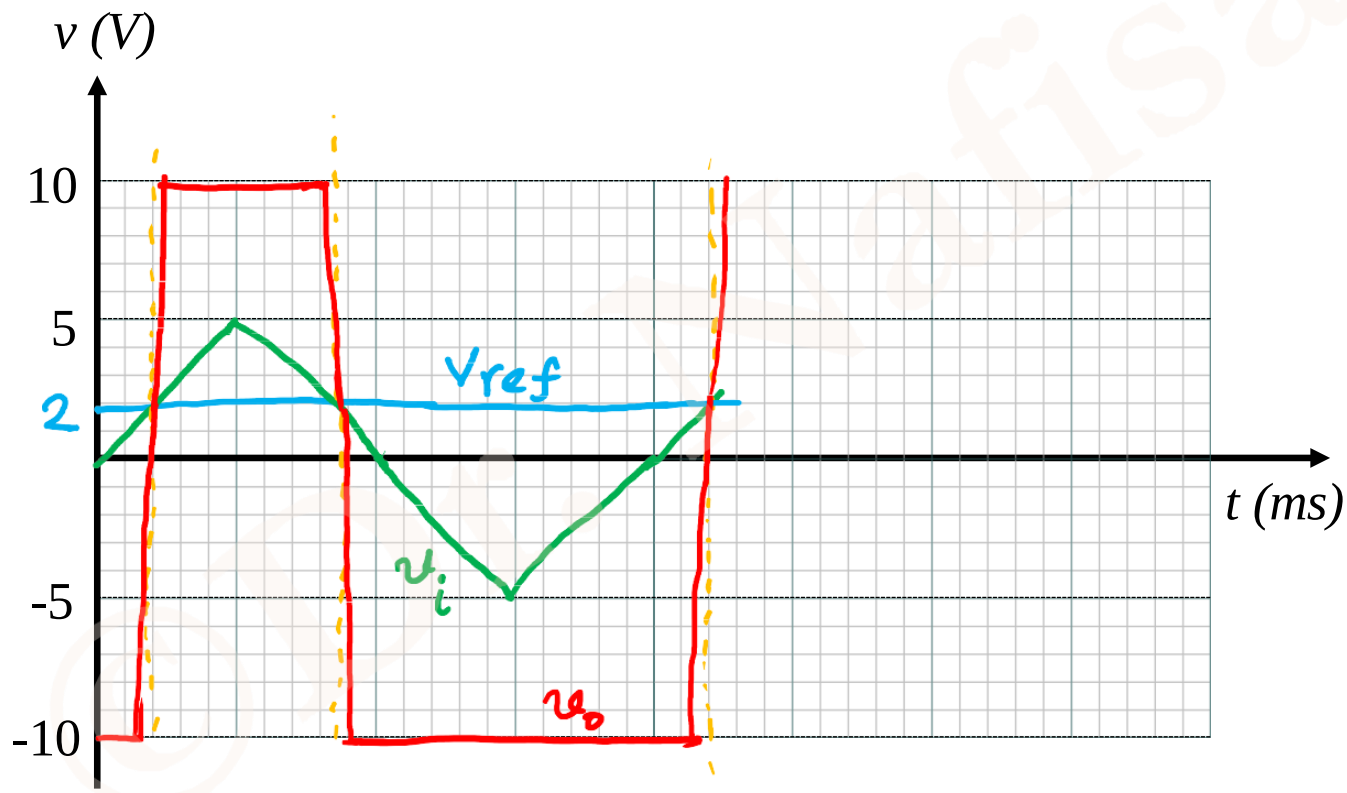
Problem 3. The input signal v_i is a triangular waveform of 1 kHz frequency and 5 V peak. The reference V_{ref} is +2 V. Draw (a) the input waveform v_i, (b) the output waveform v_o and (c) the transfer characteristics.



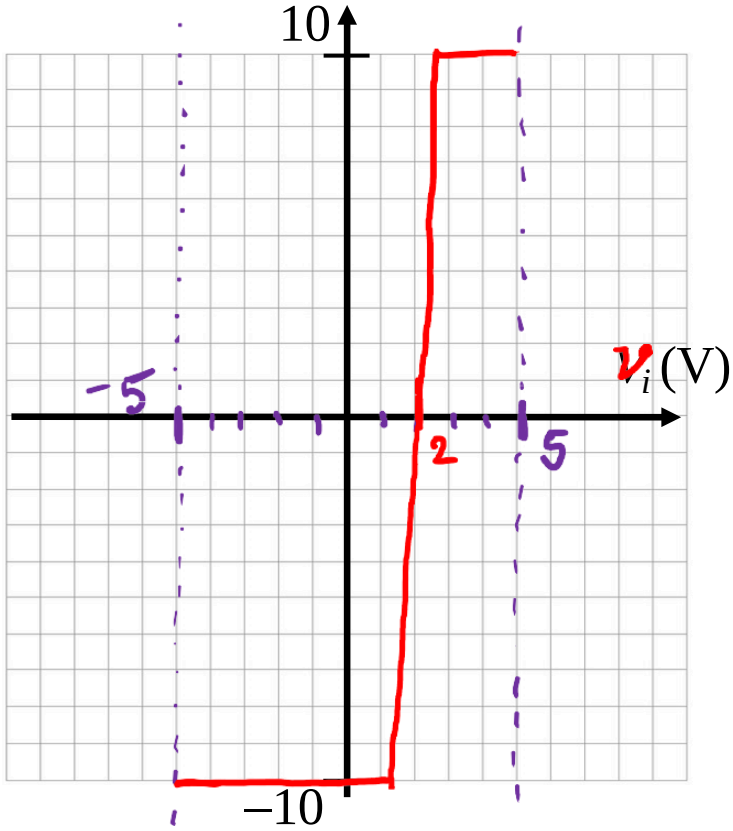
$T_+ < T_-$

open-loop comparator
Non-inv configuration

(a, b)

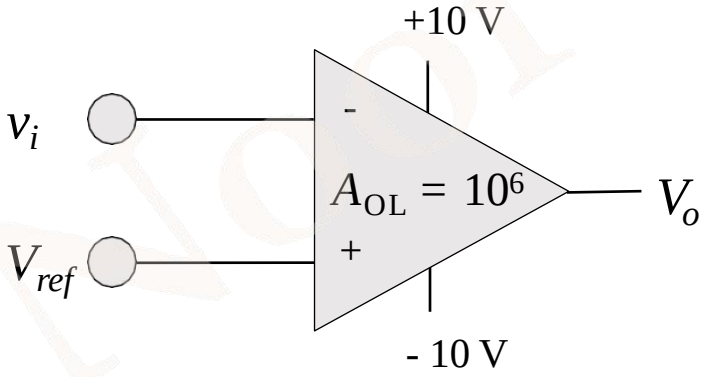


(c) v_O (V)

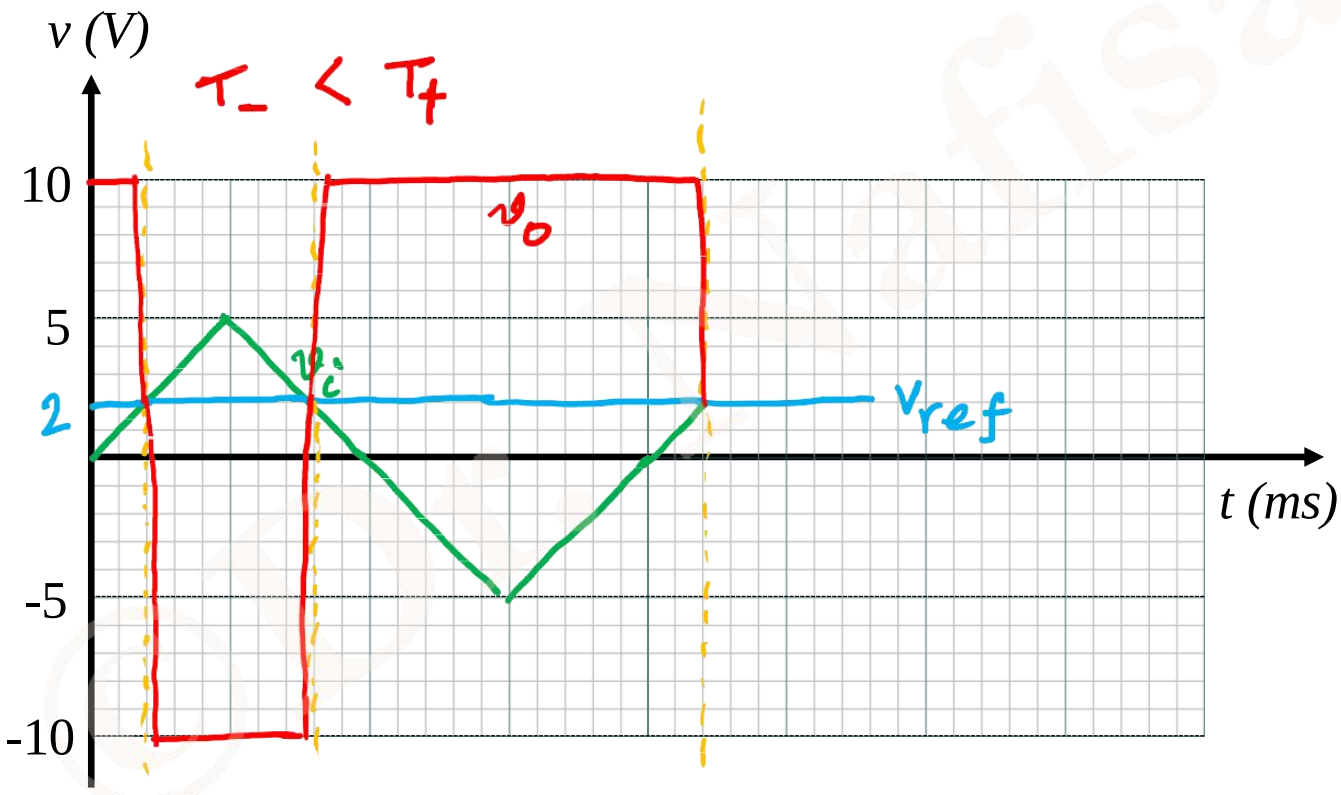


Inverting positive level detector

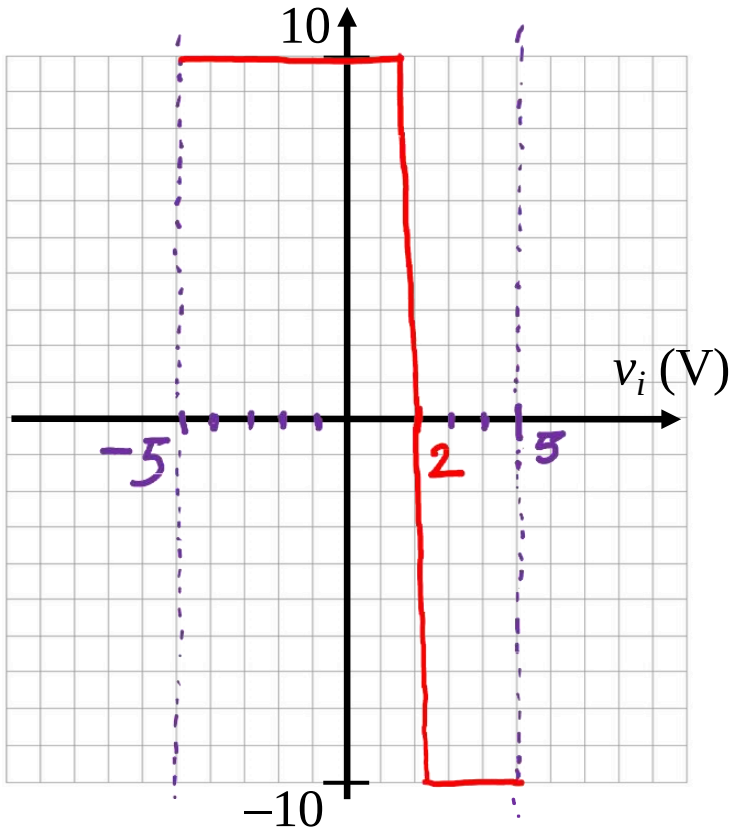
Problem 4. The input signal v_i is a triangular waveform of 1 kHz frequency and 5 V peak. The reference V_{ref} is +2 V. Draw (a) the input waveform v_i, (b) the output waveform v_o and (c) the transfer characteristics.



(a, b)

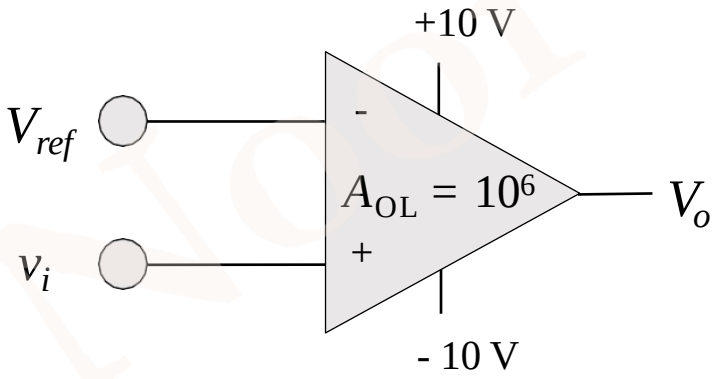


(c) v_O (V)

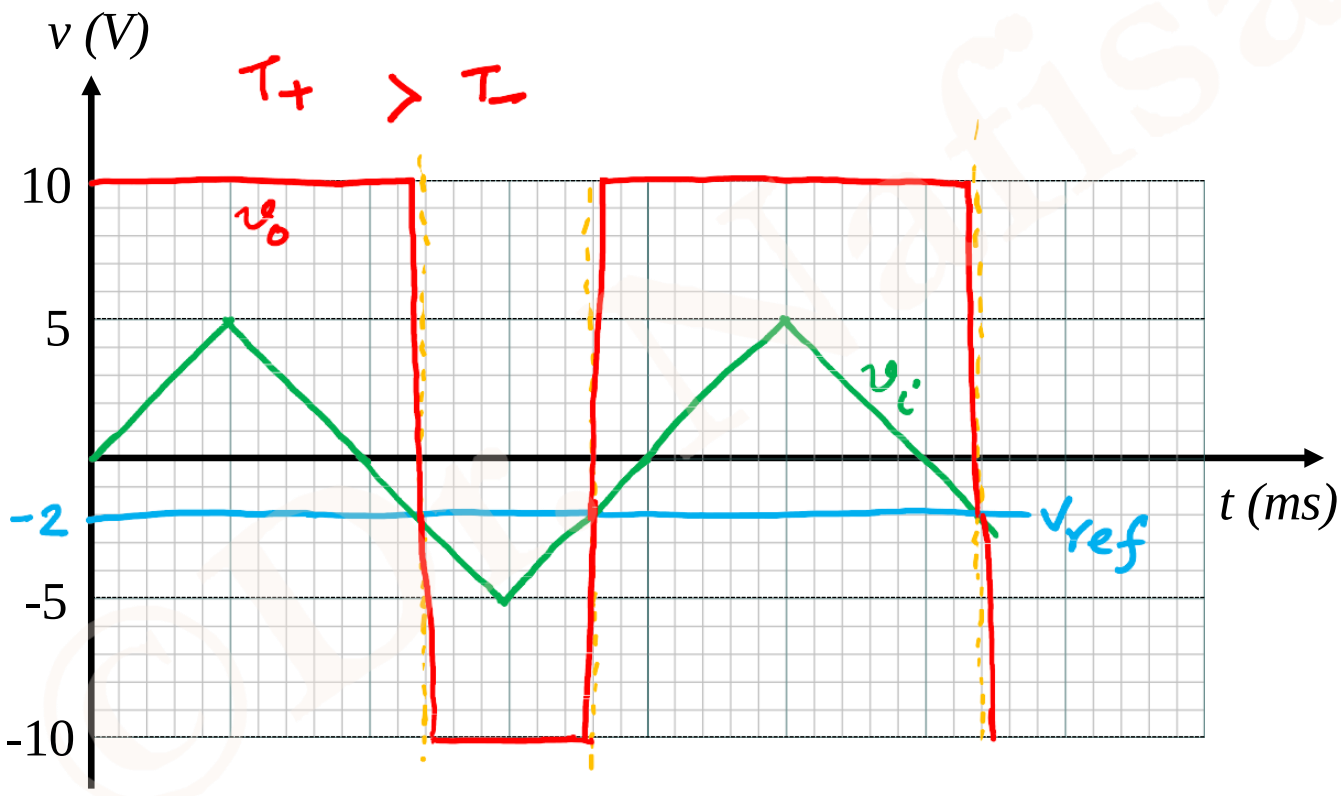


Non-inverting negative level detector

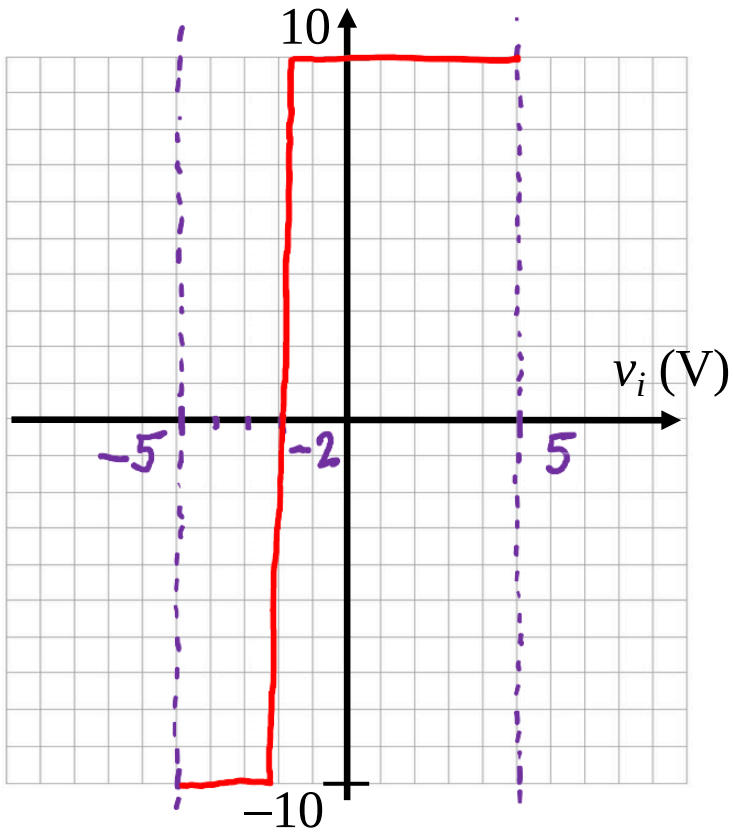
Problem 5. The input signal v_i is a triangular waveform of 1 kHz frequency and 5 V peak. The reference V_{ref} is -2 V. Draw (a) the input waveform v_i, (b) the output waveform v_o and (c) the transfer characteristics.



(a, b)

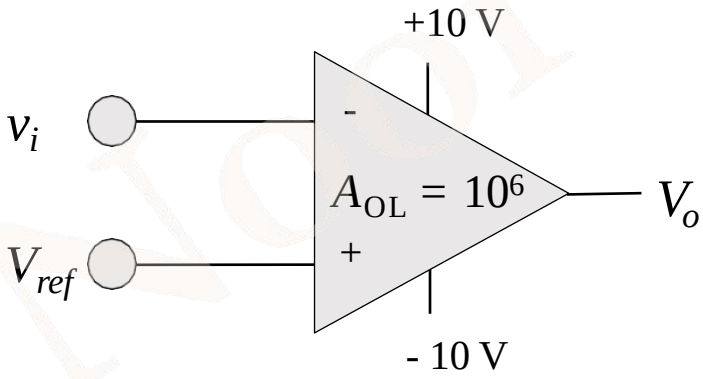


(c) v_O (V)

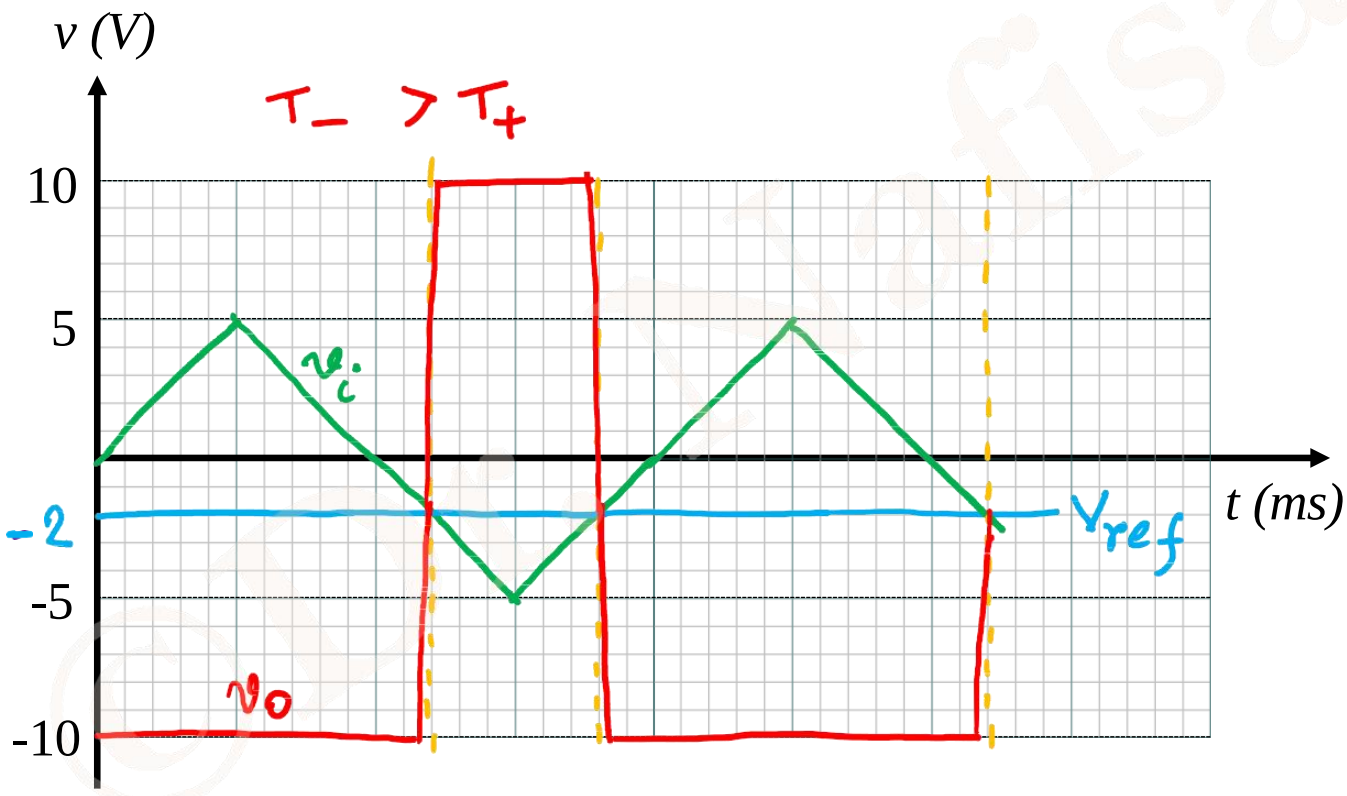


Inverting negative level detector

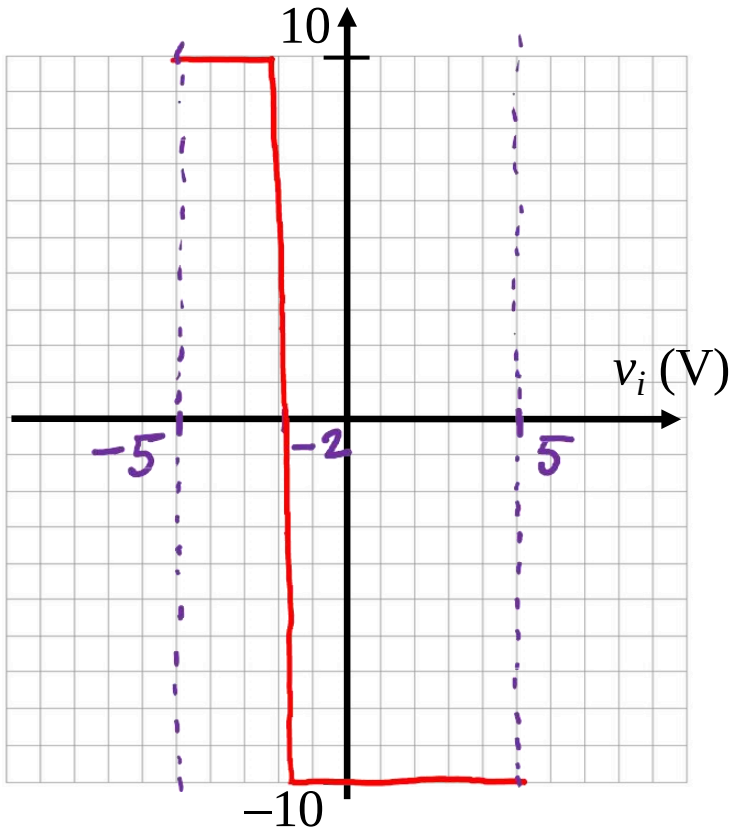
Problem 6. The input signal v_i is a triangular waveform of 1 kHz frequency and 5 V peak. The reference V_{ref} is -2 V. Draw (a) the input waveform v_i, (b) the output waveform v_o and (c) the transfer characteristics.



(a, b)

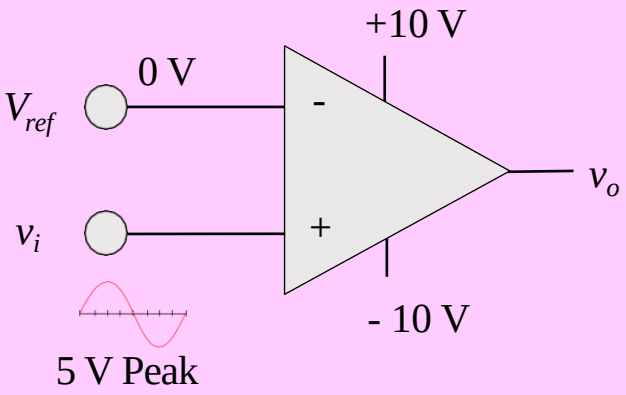


(c) v_O (V)

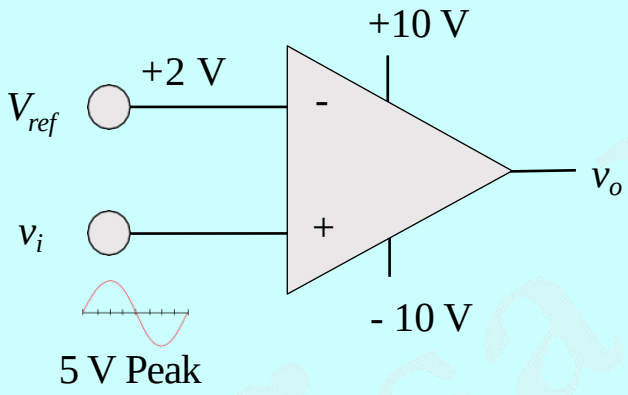


Review: Open-loop comparator circuits

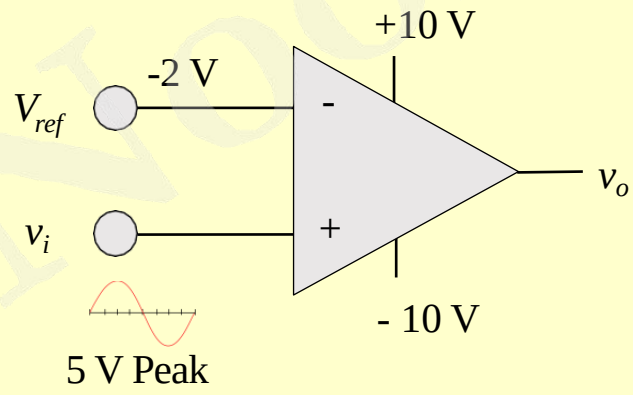
Non-inverting Zero Crossing Detector



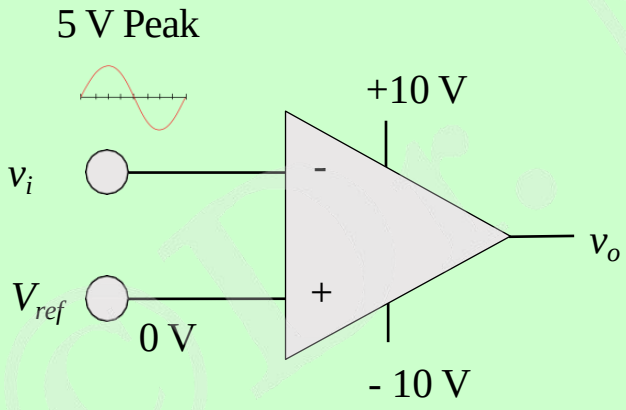
Non-inverting +ve level indicator



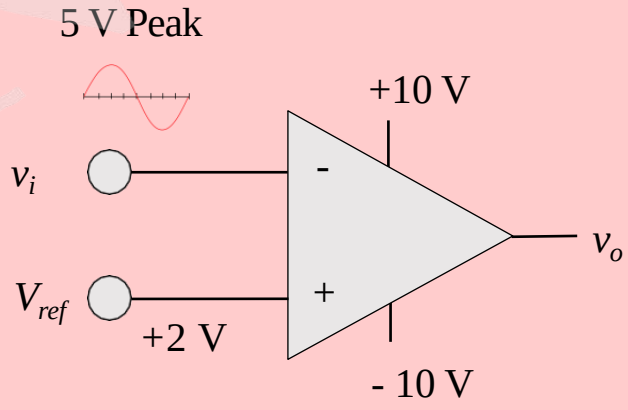
Non-inverting -ve level indicator



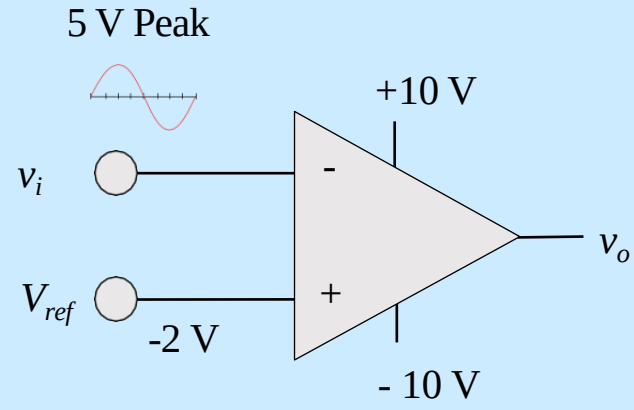
Inverting Zero Crossing Detector



Inverting +ve level indicator



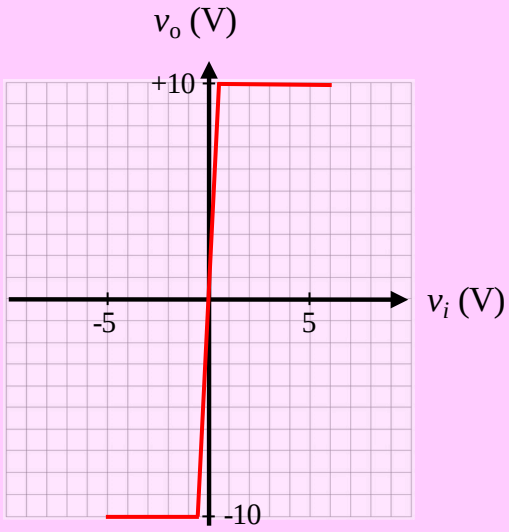
Inverting -ve level indicator



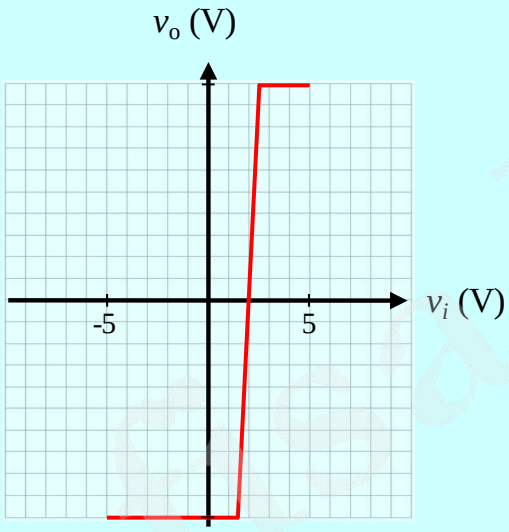
Review: Open-loop comparators

Transfer characteristics

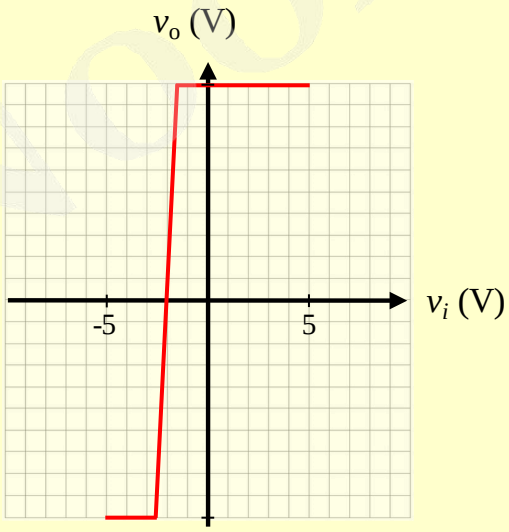
Non-inverting Zero Crossing Detector



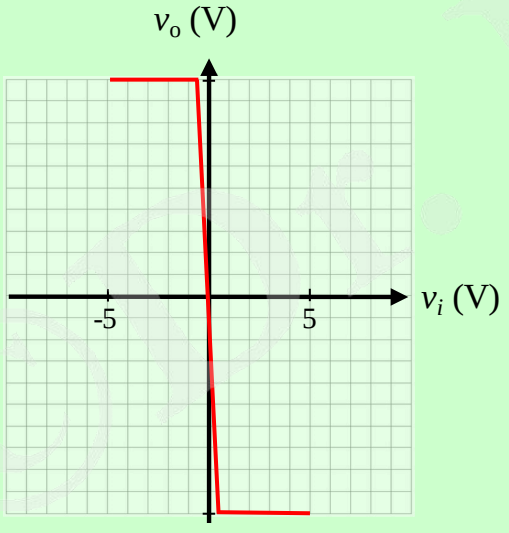
Non-inverting +ve level indicator



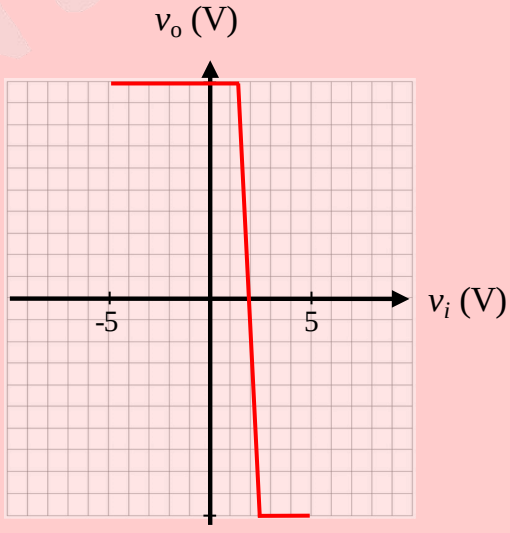
Non-inverting -ve level indicator



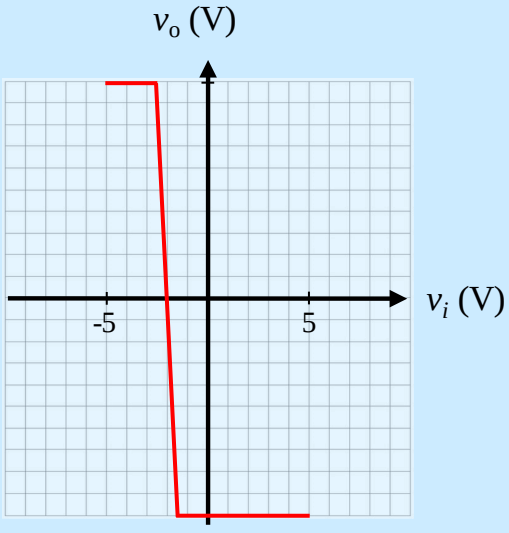
Inverting Zero Crossing Detector



Inverting +ve level indicator

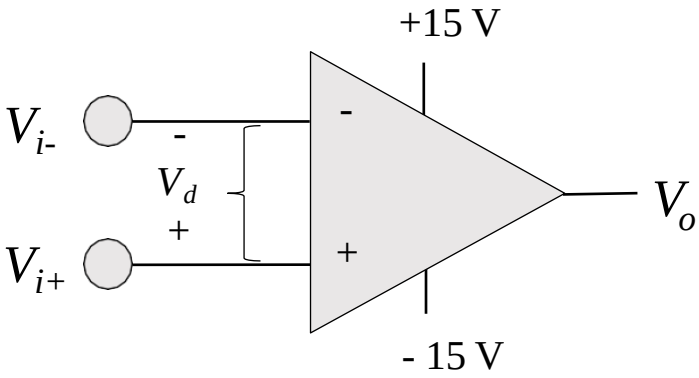


Inverting -ve level indicator



Open-loop comparator

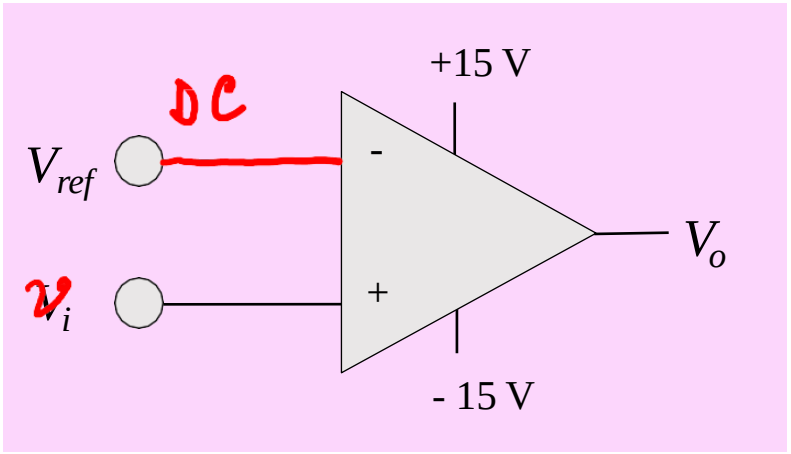
Problem 7. V_{i-} and V_{i+} are the dc inputs at the -ve and +ve input terminals, respectively. These voltage are given with respect to the ground. $A_{OL} = 2\times10^5$. $V_{sat+} = +15$ V, $V_{sat-} = -15$ V. Assume ideal conditions. Calculate the output voltage V_o for the following cases:

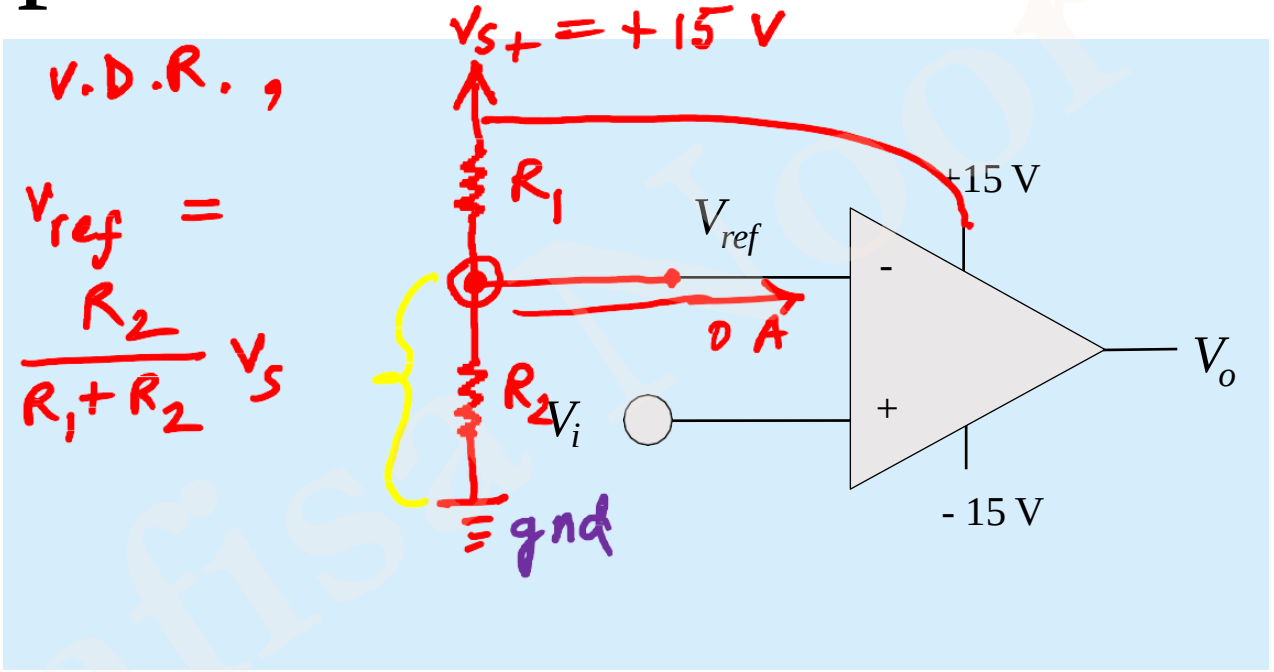


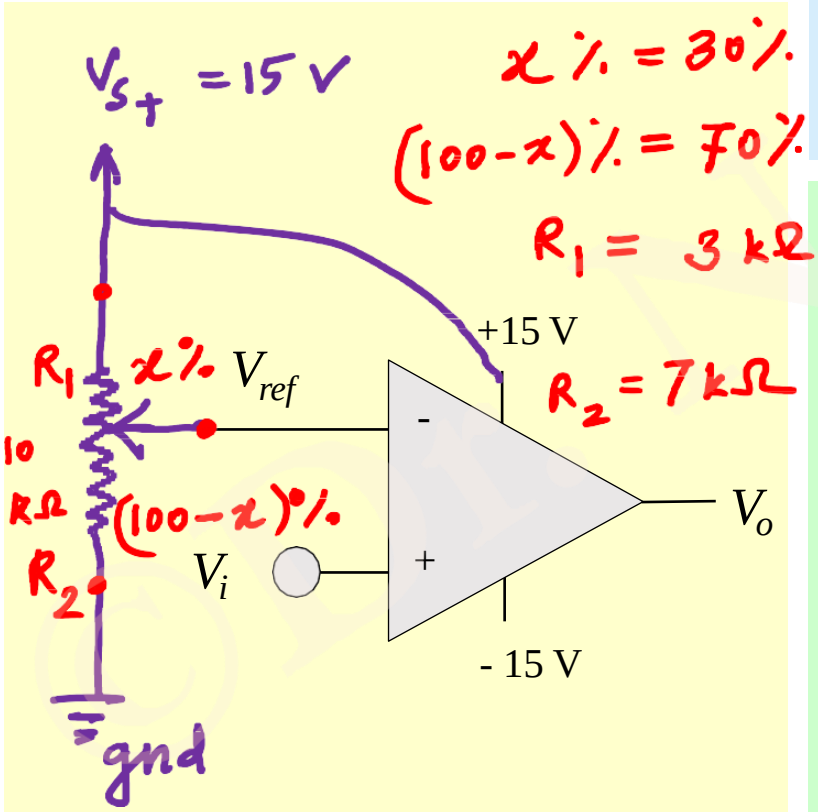
V_{i-}	V_{i+}	V_d	V_o
-10 µV	-15 µV		
-10 µV	+15 µV		
-10 µV	-5 µV		
+1.000001 V	+1.000000 V		
+5 mV	0		
0	+5 mV		

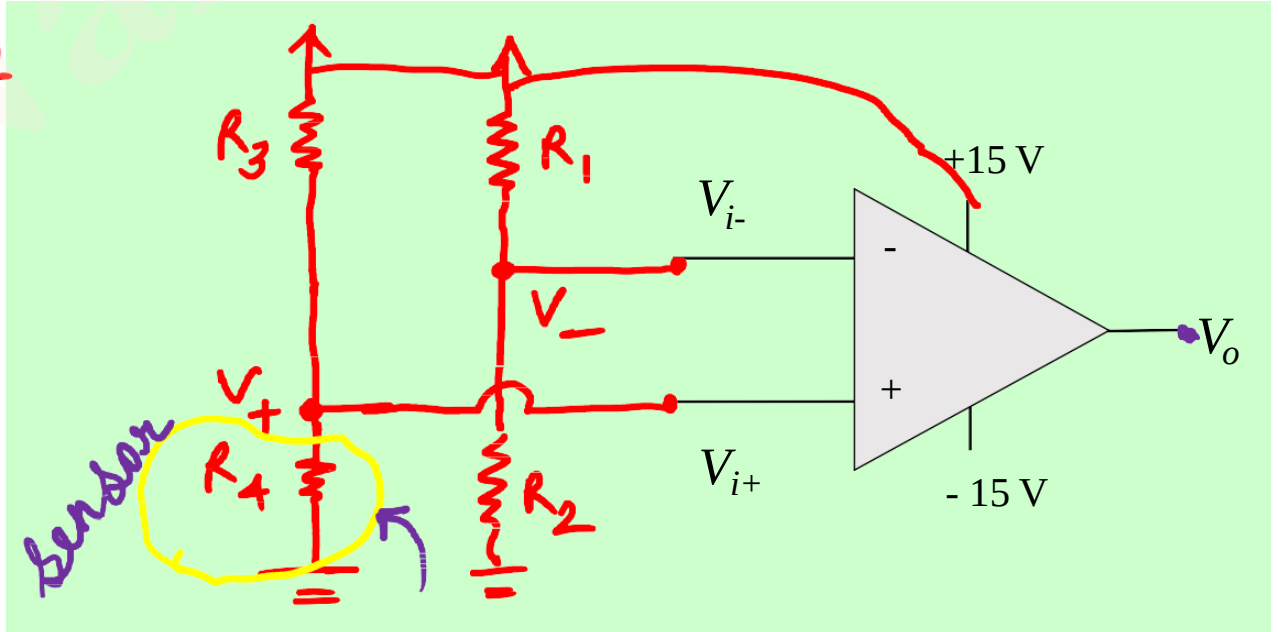
Source: Coughlin Example 2-1

Open-loop comparator - reference



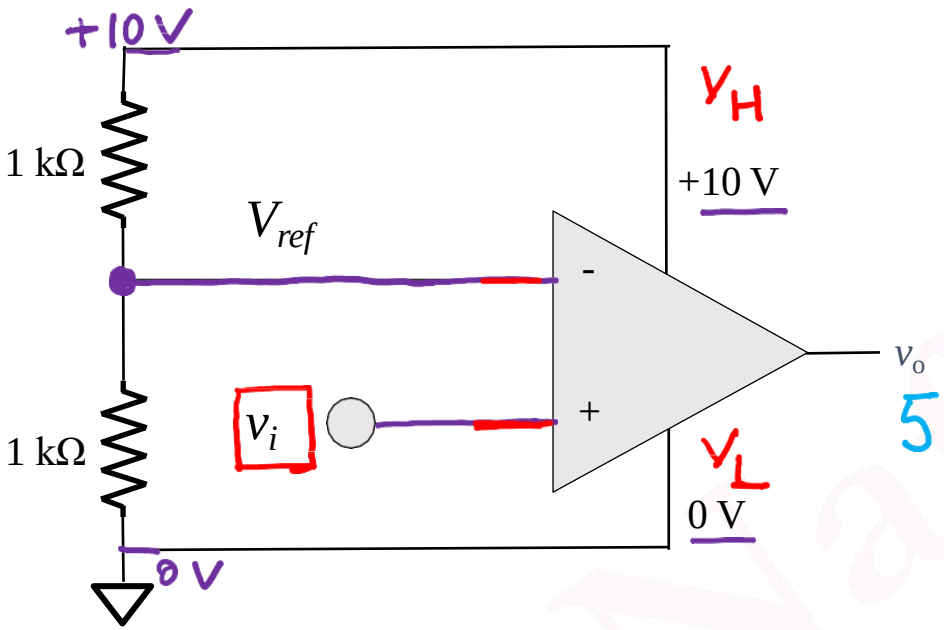






Open-loop comparator

Problem 8. For the following circuit, $A_{OL} = 10^5$. Assume ideal conditions. Calculate the output voltage v_o.



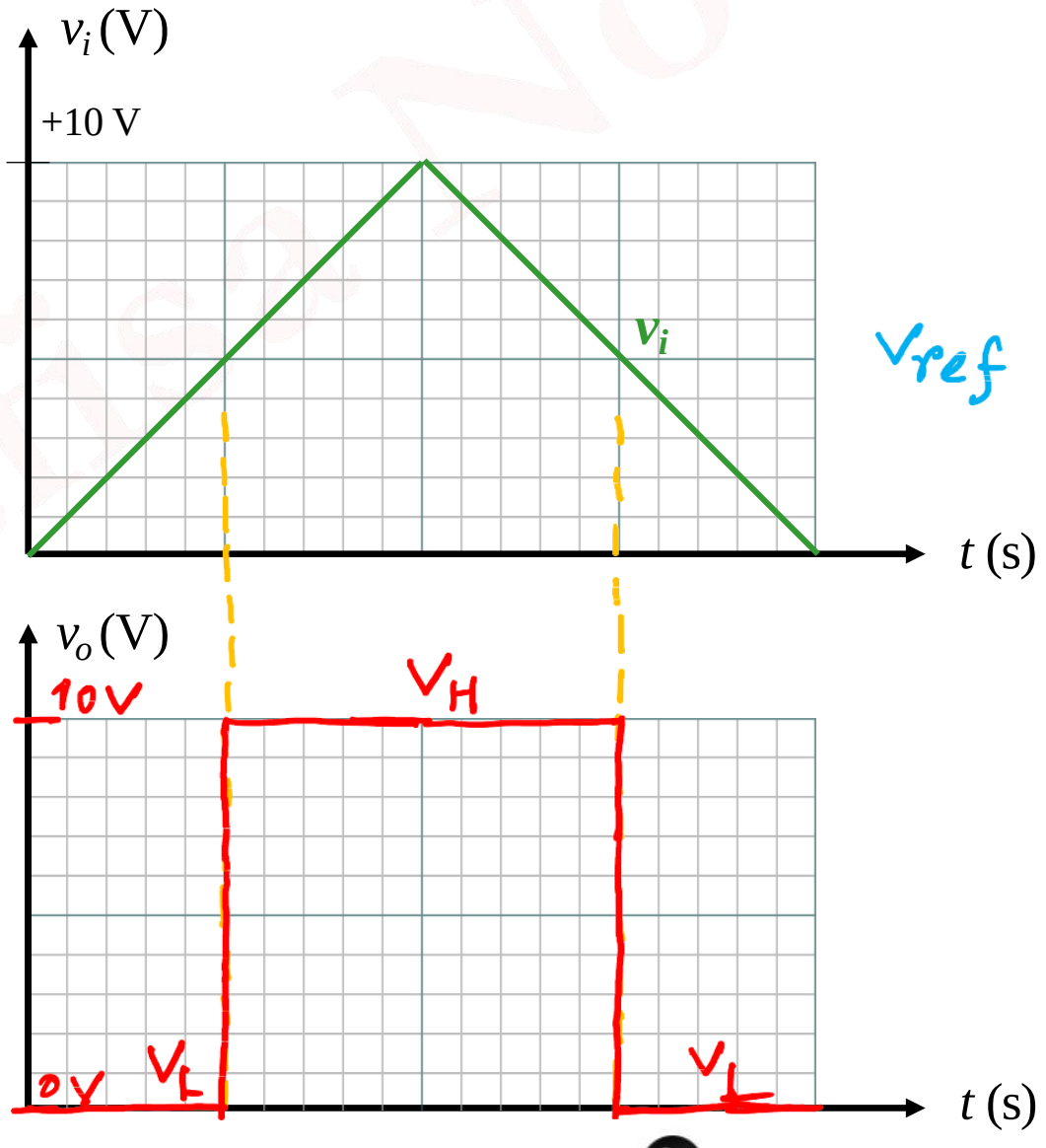
V_H

V_L

$$V_{ref} = \frac{1}{1+1} \cdot 10\ V$$

$$V_{ref} = 5\ V$$

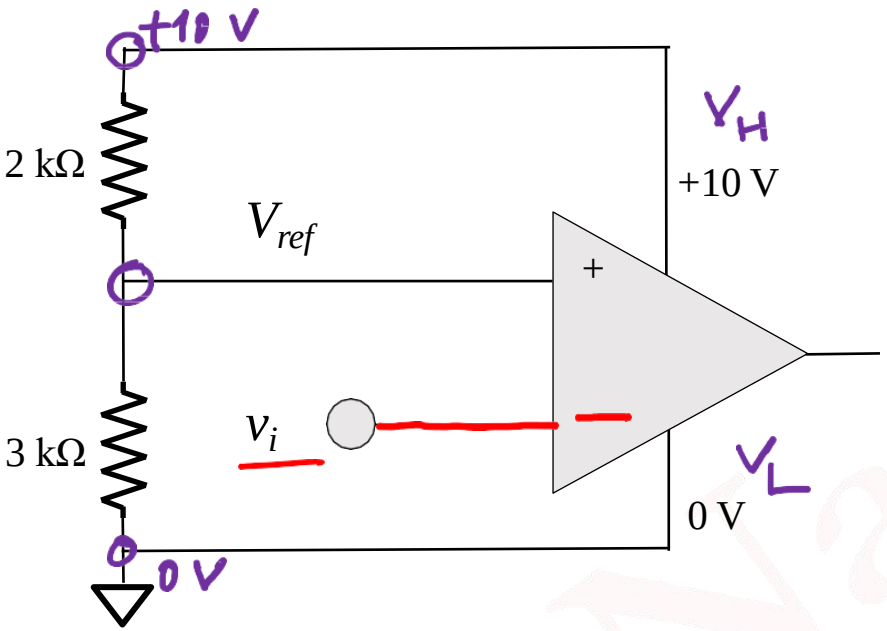
non-inv



Source: ElectronicsTutorials

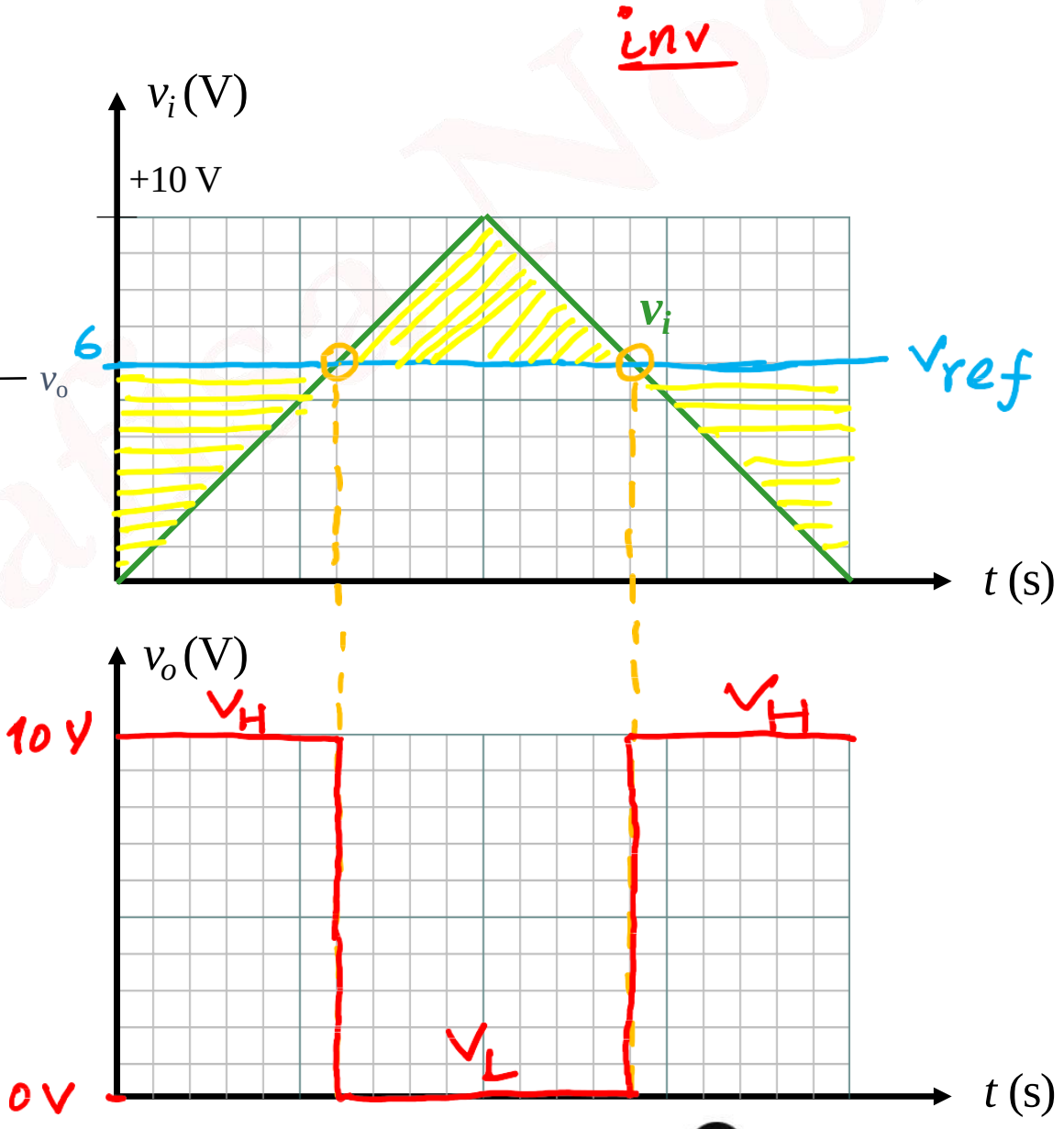
Open-loop comparator

Problem 9. For the following circuit, $A_{OL} = 10^5$. Assume ideal conditions. Calculate the output voltage v_o.



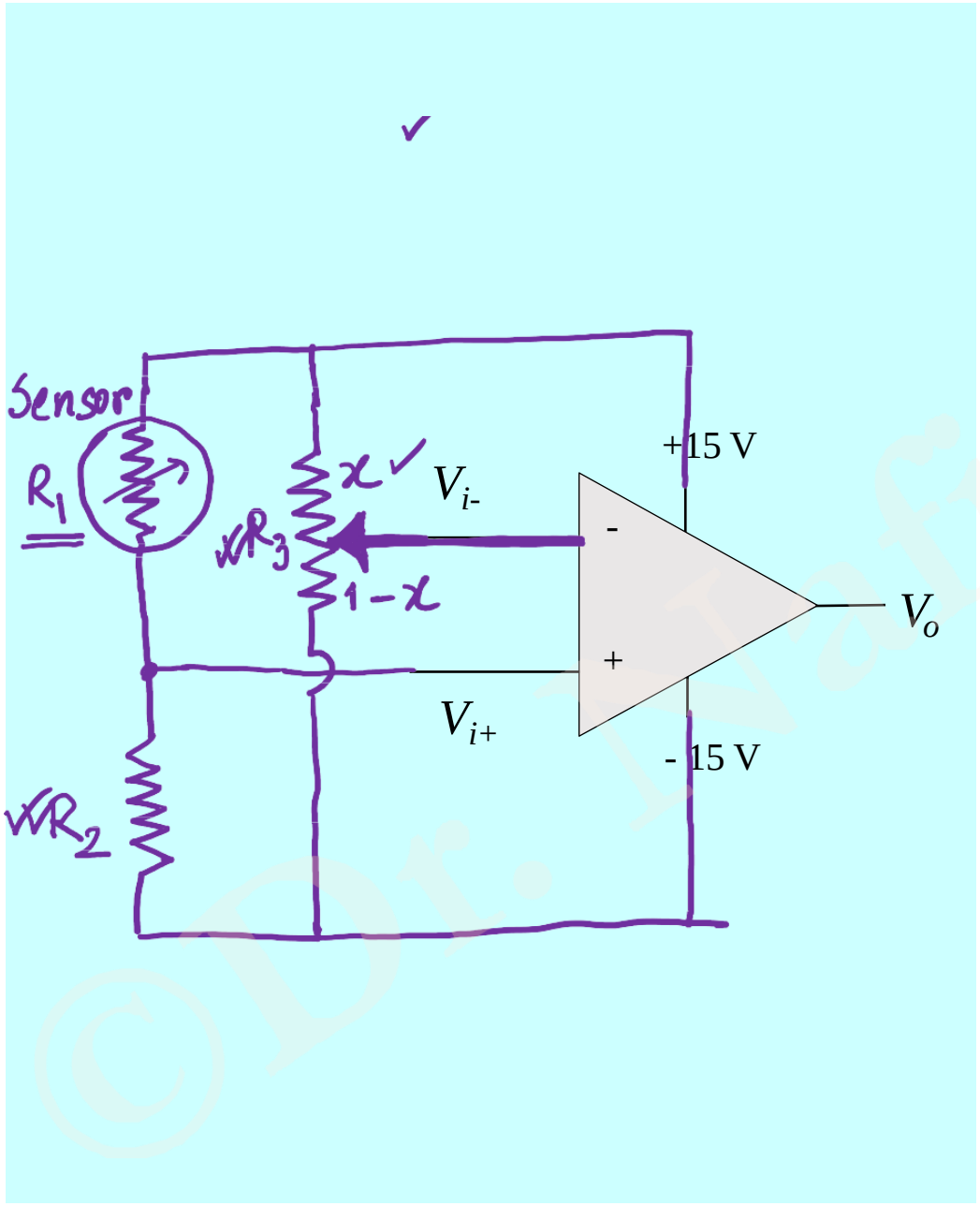
$$V_{ref} = \frac{3}{2+3} \times 10 \text{ V}$$

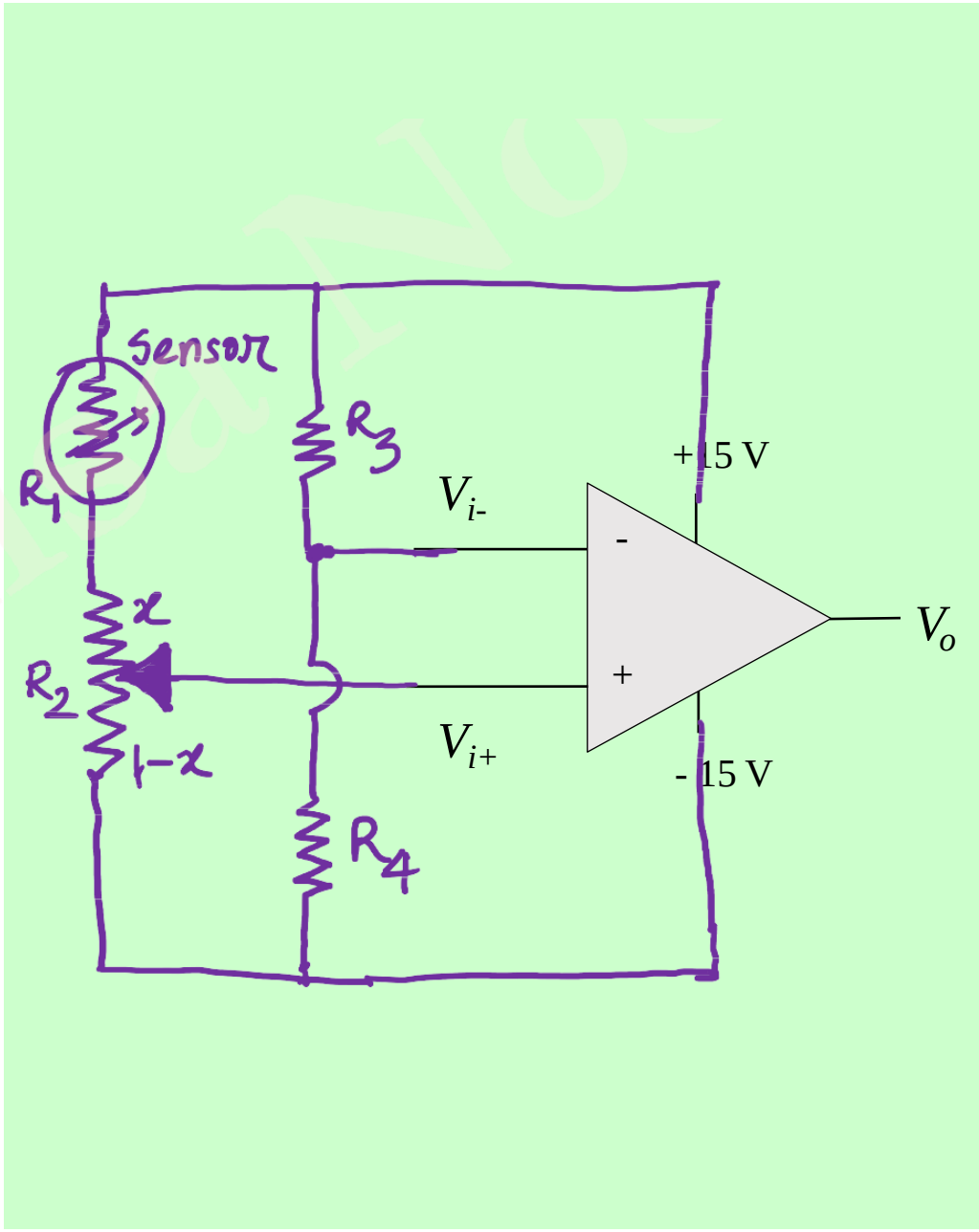
$$V_{ref} = 6 \text{ V}$$



Source: Electronics Tutorials

Open-loop comparator - applications





Light Sensor Switch Circuit

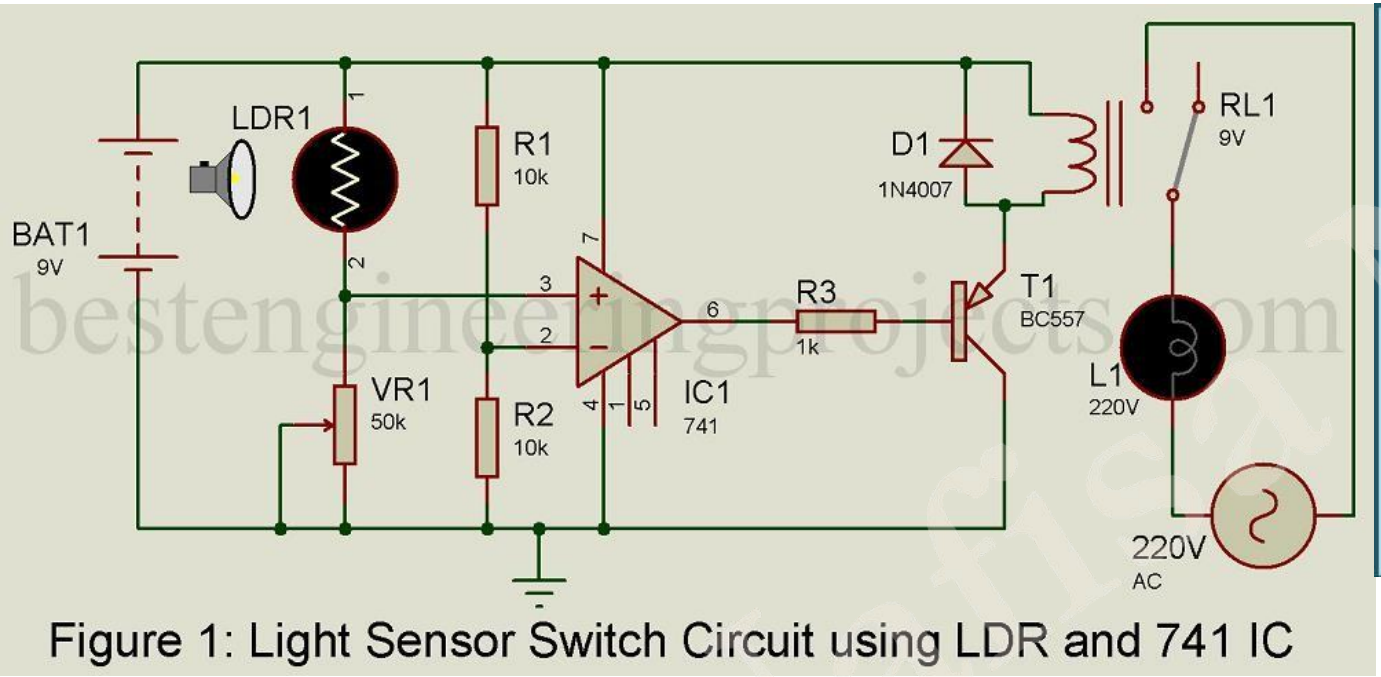
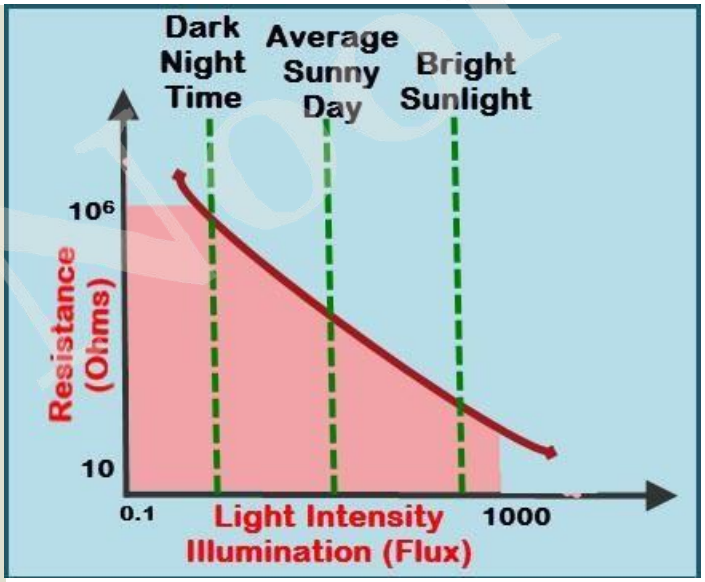


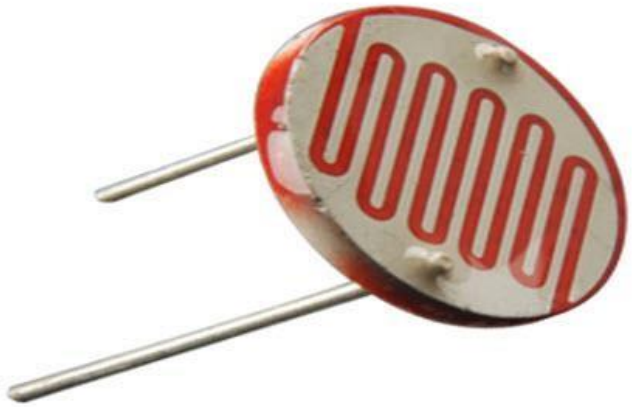
Figure 1: Light Sensor Switch Circuit using LDR and 741 IC

Source: Engineering Projects



WatElectronics.com

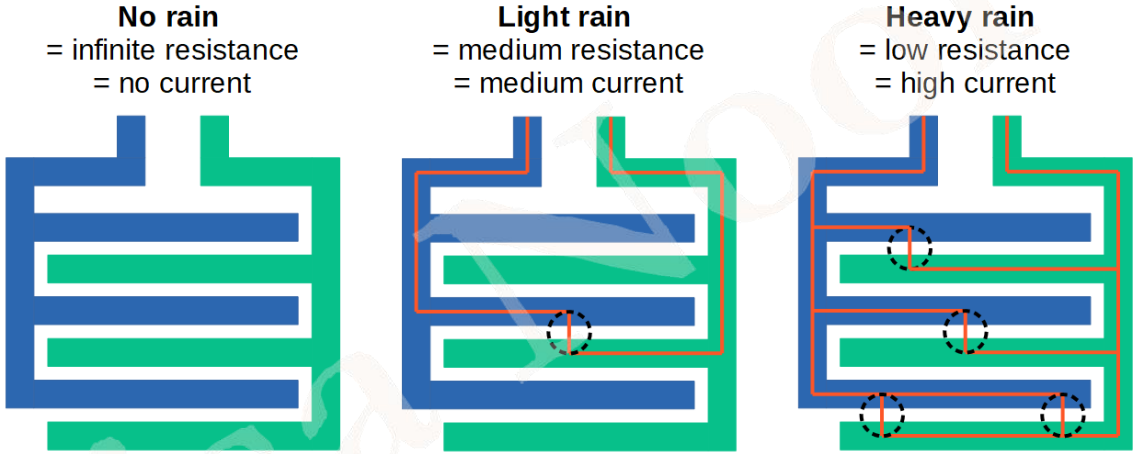
Light dependent resistor

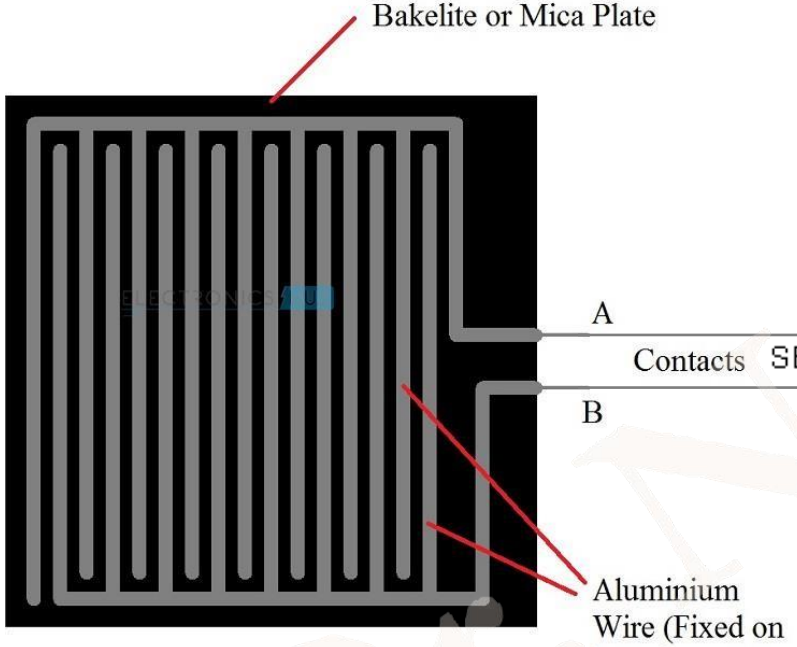


Source: EL-PRO-CUS ELECTRONICS | PROJECTS | FOCUS

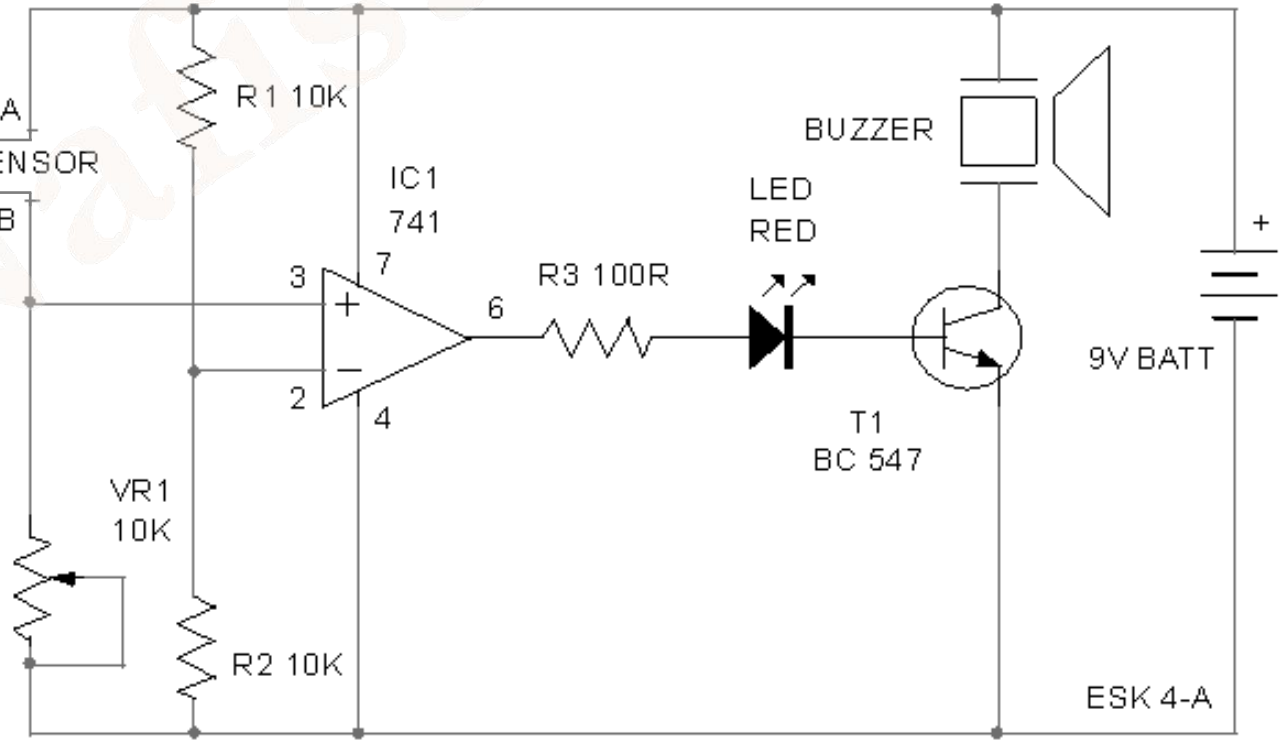
Rain detector circuit

Source: DiyIOt



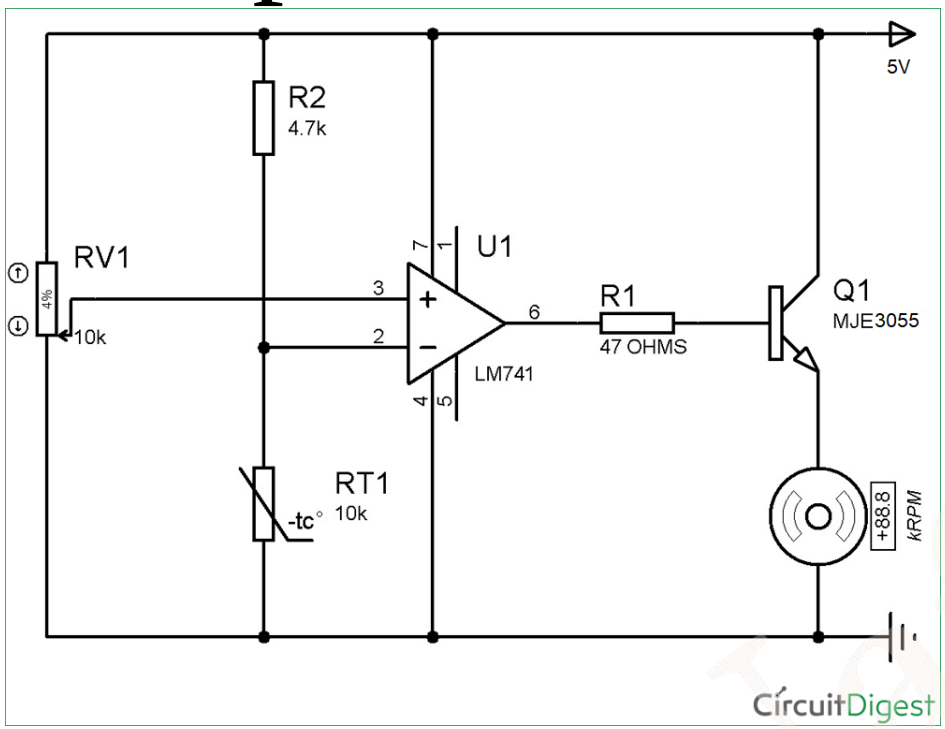


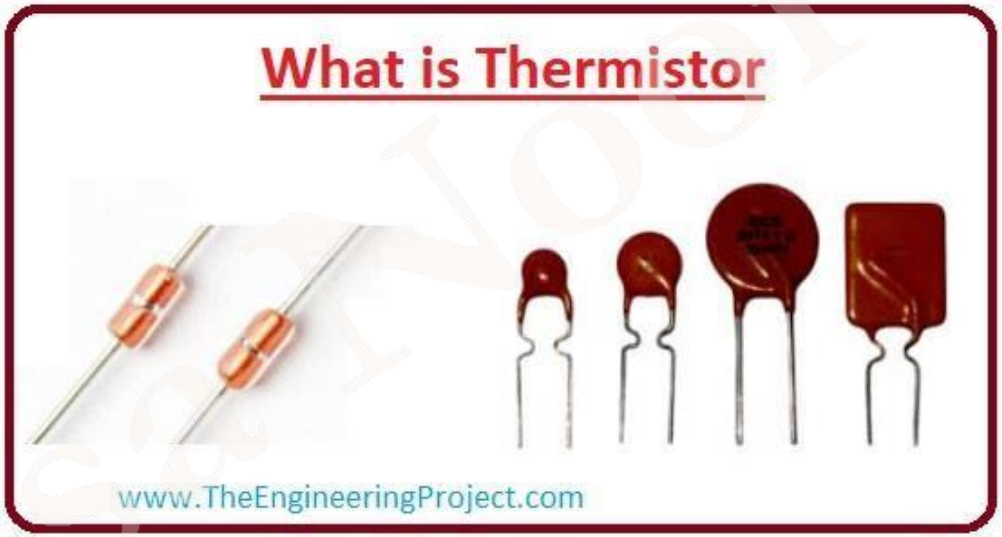
Source: ELECTRONICS HUB

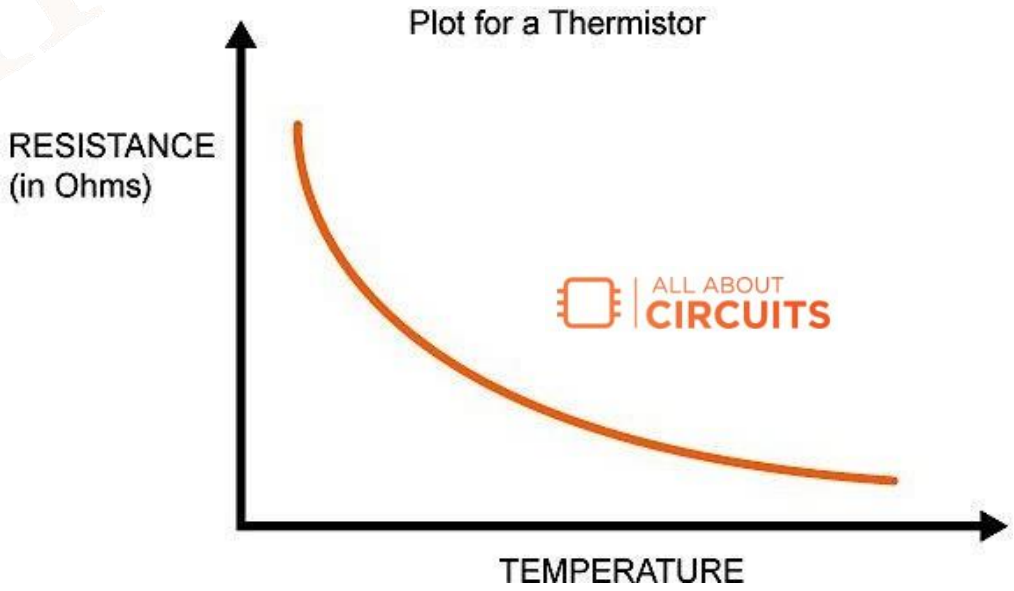


Source: Hobby Circuits and Projects

Temperature controlled DC fan

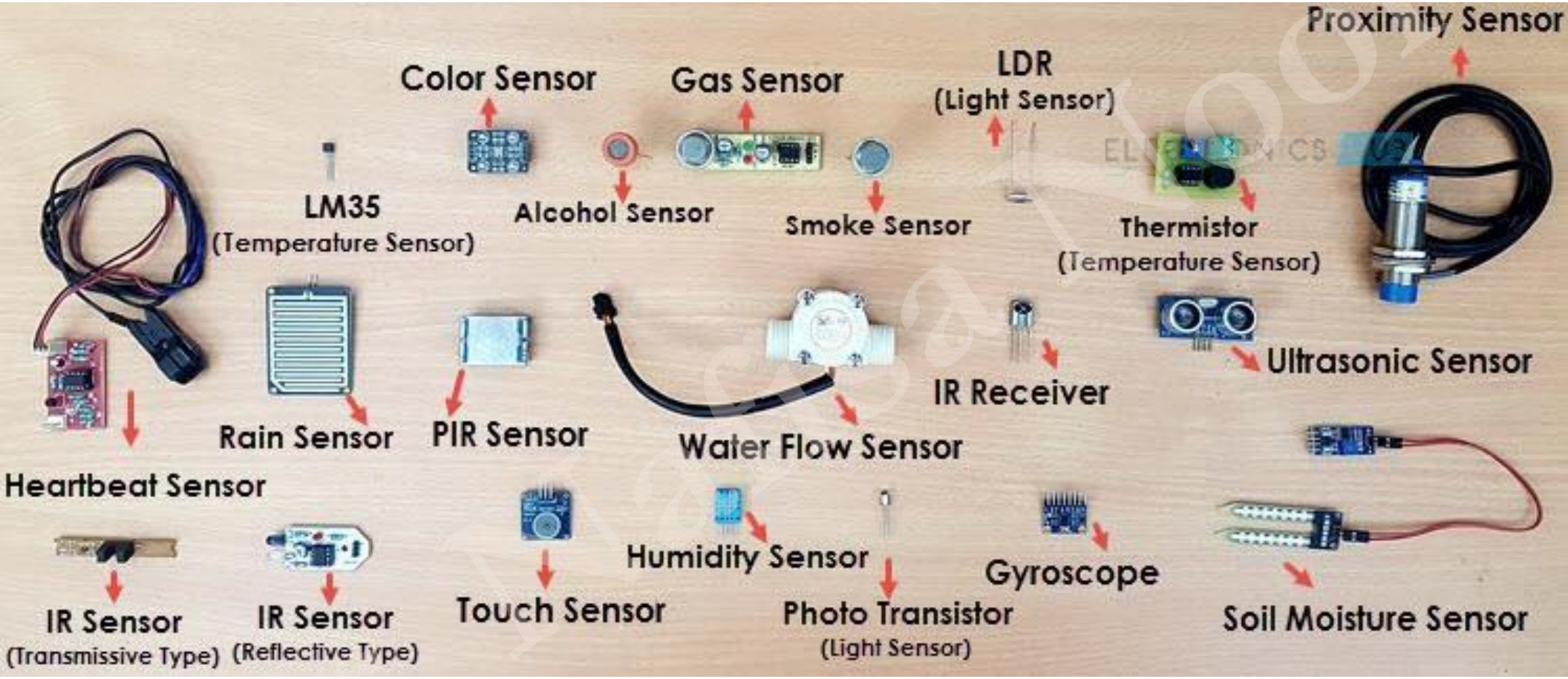






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Various types of sensors



Source: ELECTRONICS HUB